

# DACx1001 20-Bit, 18-Bit, and 16-Bit, Low-Noise, Ultra-Low Harmonic Distortion, Fast-Settling, High-Voltage Output, Digital-to-Analog Converters (DACs)

## 1 Features

- 20-bit monotonic: 1-LSB DNL (max)
- Integral linearity: 4-LSB INL (max)
- Low noise:  $7\text{ nV}/\sqrt{\text{Hz}}$
- Code independent low glitch: 1 nV-s
- Excellent THD:  $-105\text{ dB}$  at 1-kHz  $f_{\text{OUT}}$
- Fast settling: 1  $\mu\text{s}$
- Flexible output ranges:  $V_{\text{REFPF}}$  to  $V_{\text{REFNF}}$
- Integrated, precision feedback resistors
- 50-MHz, 4-wire SPI-compatible interface
  - Readback
  - Daisy-chain
- Temperature range:  $-40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$
- Package: 48-pin TQFP

## 2 Applications

- [Lab and field instrumentation](#)
- [Spectrometer](#)
- [Analog output module](#)
- [Battery Test](#)
- [Semiconductor test](#)
- [Arbitrary waveform generator \(AWG\)](#)
- [MRI](#)
- [X-ray systems](#)
- [Professional audio amplifier \(rack mount\)](#)

## 3 Description

The 20-bit DAC11001A, 18-bit DAC91001, and 16-bit DAC81001 (DACx1001) are highly accurate, low-noise, voltage-output, single-channel, digital-to-analog converters (DACs). The DACx1001 are specified monotonic by design, and offer excellent linearity of less than 4 LSB (max) across all ranges.

The unbuffered voltage output offers low noise performance ( $7\text{ nV}/\sqrt{\text{Hz}}$ ) in combination with a fast settling time (1  $\mu\text{s}$ ), making this device an excellent choice for low-noise, fast control-loop, and waveform generation applications. The DACx1001 integrates an enhanced deglitch circuit with code-independent ultra-low glitch (1 nV-s) to enable clean waveform ramps with ultra-low total harmonic distortion (THD).

The DACx1001 devices incorporate a power-on-reset circuit so that the DAC powers with known values in the registers. With external references, DAC output ranges from  $V_{\text{REFPF}}$  to  $V_{\text{REFNF}}$  can be achieved, including asymmetric output ranges.

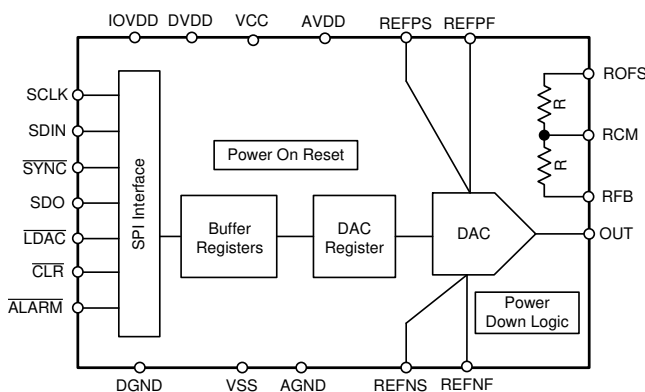
The DACx1001 use a versatile 4-wire serial interface that operates at clock rates of up to 50 MHz. The DACx1001 is specified over the industrial temperature range of  $-40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ .

### Device Information<sup>(1)</sup>

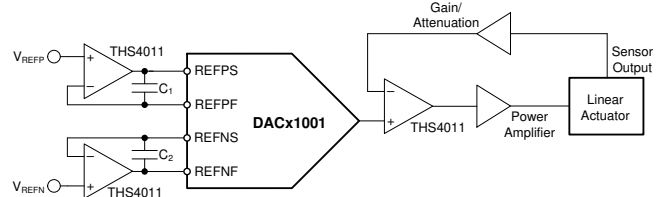
| PART NUMBER | PACKAGE   | BODY SIZE (NOM)   |
|-------------|-----------|-------------------|
| DAC11001A   | TQFP (48) | 7.00 mm × 7.00 mm |
| DAC91001    |           |                   |
| DAC81001    |           |                   |

(1) For all available packages, see the package option addendum at the end of the data sheet.

### Functional Block Diagram



### High-Precision, Control-Loop Circuit



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## 4 Revision History

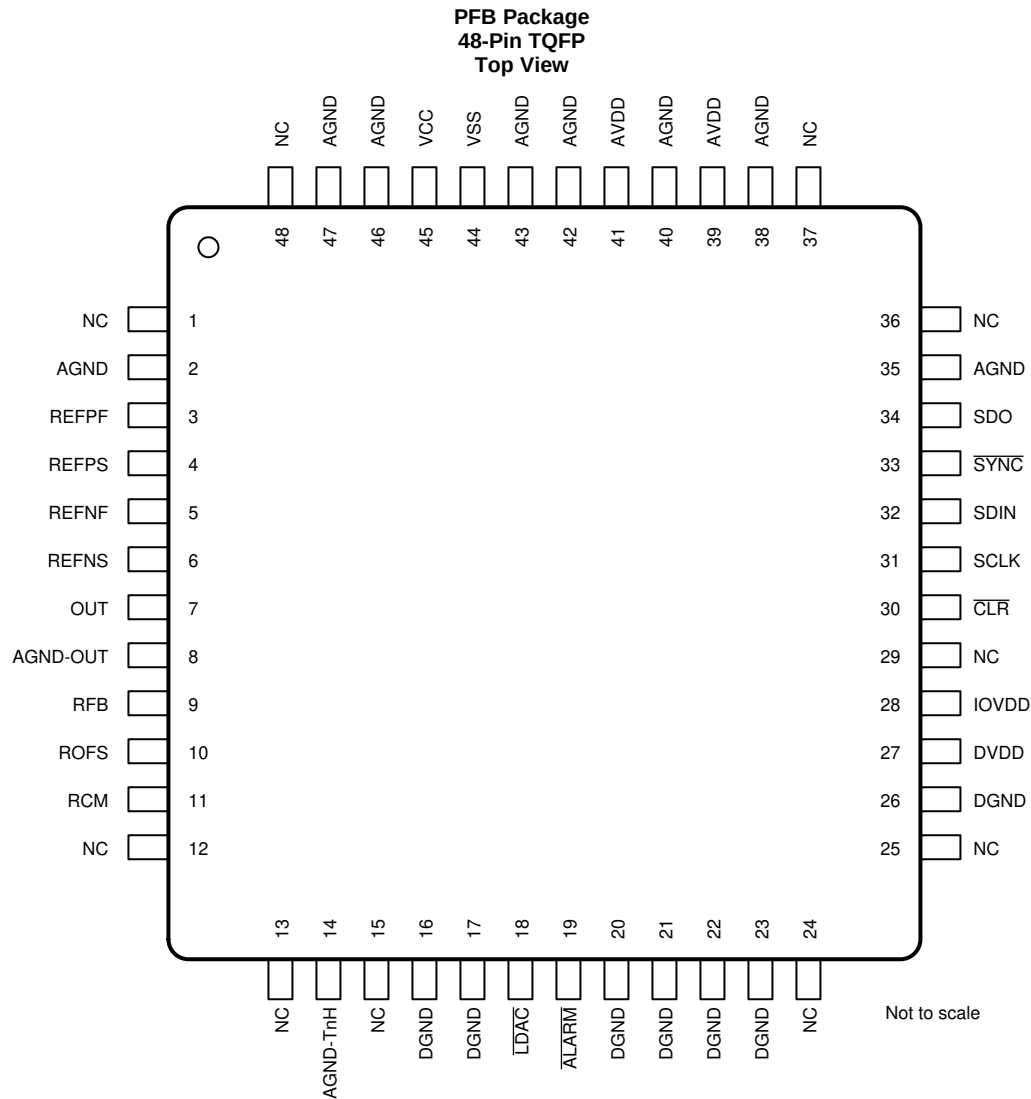
| Changes from Revision A (December 2019) to Revision B                                                                              | Page |
|------------------------------------------------------------------------------------------------------------------------------------|------|
| • Changed DAC81001 and DAC91001 from advanced information (preview) to production data (active), and added associated content      | 1    |
| • Changed relative accuracy drift over time typical value from 0.1 LSB to $\pm 0.1$ LSB in <i>Electrical Characteristics</i> table | 7    |
| • Added output voltage drift over time parameter to the <i>Electrical Characteristics</i> table                                    | 8    |
| • Changed Figure 42, DAC Output Noise: 0.1 Hz to 10 Hz                                                                             | 22   |

| Changes from Original (October 2019) to Revision A                                         | Page |
|--------------------------------------------------------------------------------------------|------|
| • Changed DAC11001A device from advanced information (preview) to production data (active) | 1    |

## 5 Device Comparison Table

| DEVICE    | RESOLUTION |
|-----------|------------|
| DAC11001A | 20-bit     |
| DAC91001  | 18-bit     |
| DAC81001  | 16-bit     |

## 6 Pin Configuration and Functions



## Pin Functions

| PIN      |                                       | TYPE           | DESCRIPTION                                                                                                                                                                      |
|----------|---------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME     | NO.                                   |                |                                                                                                                                                                                  |
| AGND     | 2, 35, 38, 40, 42, 43, 46, 47         | Analog ground  | Connect to 0 V.                                                                                                                                                                  |
| AGND-OUT | 8                                     | Analog ground  | Connect to 0 V. Measure DAC output voltage with respect to this node.                                                                                                            |
| AGND-TnH | 14                                    | Analog ground  | Connect to 0 V. Integrated deglitcher clock ground..                                                                                                                             |
| ALARM    | 19                                    | Output         | Alarm output                                                                                                                                                                     |
| AVDD     | 39, 41                                | Power          | Positive low voltage analog power supply                                                                                                                                         |
| CLR      | 30                                    | Input          | DAC registers clear pin, active low                                                                                                                                              |
| DGND     | 16, 17, 20, 21, 22, 23, 26            | Digital ground | Connect to 0 V.                                                                                                                                                                  |
| DVDD     | 27                                    | Power          | Digital power supply pin                                                                                                                                                         |
| RFB      | 9                                     | Input          | Integrated precision resistor feedback node                                                                                                                                      |
| IOVDD    | 28                                    | Power          | Interface power supply pin                                                                                                                                                       |
| LDAC     | 18                                    | Input          | Load DAC pin, active low                                                                                                                                                         |
| NC       | 1, 12, 13, 15, 24, 25, 29, 36, 37, 48 | —              | No connection, leave floating                                                                                                                                                    |
| OUT      | 7                                     | Output         | Unbuffered voltage output                                                                                                                                                        |
| RCM      | 11                                    | Input          | Integrated precision resistor common-mode node                                                                                                                                   |
| REFNF    | 5                                     | Input          | External negative reference input. Connect to 0 V for unipolar DAC output.                                                                                                       |
| REFNS    | 6                                     | Input          | External negative reference sense node                                                                                                                                           |
| REFPF    | 3                                     | Input          | External positive reference input                                                                                                                                                |
| REFPS    | 4                                     | Input          | External positive reference sense node                                                                                                                                           |
| ROFS     | 10                                    | Input          | Integrated precision resistor offset node                                                                                                                                        |
| SCLK     | 31                                    | Input          | Serial clock input of serial peripheral interface (SPI). Schmitt-trigger logic input. Data are transferred at rates of up to 50 MHz.                                             |
| SDIN     | 32                                    | Input          | Serial data input. Schmitt-trigger logic input. Data are clocked into the input shift register on the falling edge of the serial clock input.                                    |
| SDO      | 34                                    | Output         | Serial data output. Data are valid on the falling edge of SCLK.                                                                                                                  |
| SYNC     | 33                                    | Input          | SPI bus chip select input (active low). Data bits are not clocked into the serial shift register unless SYNC is low. When SYNC is high, the SDO pin is in high-impedance status. |
| VCC      | 45                                    | Power          | Analog positive power supply                                                                                                                                                     |
| VSS      | 44                                    | Power          | Analog negative power supply                                                                                                                                                     |

## 7 Specifications

### 7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)<sup>(1)</sup>

|           |                                                      |                                    | MIN            | MAX              | UNIT |
|-----------|------------------------------------------------------|------------------------------------|----------------|------------------|------|
|           | Positive supply voltage                              | $AV_{DD}$ to AGND                  | –0.3           | 7                | V    |
|           |                                                      | $V_{CC}$ to $V_{SS}$               | –0.3           | 40               |      |
|           |                                                      | $V_{CC}$ to AGND                   | –0.3           | 40               |      |
|           | Negative supply voltage                              | $V_{SS}$ to AGND                   | –19            | 0.3              | V    |
|           | Positive reference voltage                           | $V_{REFPF}$ to $V_{REFNF}$         | –0.3           | 40               | V    |
|           |                                                      | $V_{REFPF}$ to $V_{CC}$            | –0.3           | $V_{CC} + 0.3$   |      |
|           |                                                      | $V_{REFPF}$ to AGND                | –0.3           | 40               |      |
|           | Negative reference voltage                           | $V_{REFNF}$ to AGND                | –19            | 0.3              | V    |
|           |                                                      | $V_{REFNF}$ to $V_{SS}$            | $V_{SS} - 0.3$ | 0.3              |      |
|           | Digital and IO power supply                          | $DV_{DD}$ , $IOV_{DD}$ to DGND     | –0.3           | 7                | V    |
|           | Digital input(s) to DGND                             |                                    | DGND – 0.3     | $IOV_{DD} + 0.3$ | V    |
|           | $V_{OUT}$ , $V_{RFB}$ , $V_{RCM}$ , $V_{ROFS}$       | to AGND ( $V_{SS} = \text{AGND}$ ) | $V_{SS}$       | $V_{CC}$         | V    |
|           |                                                      | to $V_{SS}$                        | 0              | $V_{CC}$         |      |
|           | Alarm pin voltage, $\overline{\text{ALARM}}$ to DGND |                                    | –0.3           | $DV_{DD} + 0.3$  | V    |
|           | Digital output, SDO to DGND                          |                                    | –0.3           | $DV_{DD} + 0.3$  | V    |
|           | Current into any pin                                 |                                    | –10            | 10               | mA   |
| $T_J$     | Junction temperature                                 |                                    |                | 150              | °C   |
| $T_{stg}$ | Storage temperature                                  |                                    | –65            | 150              | °C   |

- (1) Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under Recommended Operating Conditions. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

### 7.2 ESD Ratings

|             |                         |                                                                                          | VALUE | UNIT |
|-------------|-------------------------|------------------------------------------------------------------------------------------|-------|------|
| $V_{(ESD)}$ | Electrostatic discharge | Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins <sup>(1)</sup>              | ±1000 | V    |
|             |                         | Charged device model (CDM), per JEDEC specification JESD22-C101, all pins <sup>(2)</sup> | ±250  |      |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.  
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

|                |                                            | MIN                     | NOM | MAX | UNIT |
|----------------|--------------------------------------------|-------------------------|-----|-----|------|
|                | AV <sub>DD</sub> to AGND                   | 4.5                     |     | 5.5 | V    |
|                | V <sub>SS</sub> to AGND                    | –18                     |     | –3  | V    |
|                | V <sub>CC</sub> to AGND                    | 8                       |     | 33  | V    |
|                | V <sub>CC</sub> to V <sub>SS</sub>         | 11                      |     | 36  | V    |
|                | DV <sub>DD</sub> to DGND                   | 2.7                     |     | 5.5 | V    |
|                | IOV <sub>DD</sub> to DGND                  | 1.7                     |     | 5.5 | V    |
|                | AGND to DGND                               | –0.3                    |     | 0.3 | V    |
|                | V <sub>IH</sub> digital input high voltage | 0.7 × IOV <sub>DD</sub> |     |     | V    |
|                | V <sub>IL</sub> digital input low voltage  | 0.3 × IOV <sub>DD</sub> |     |     | V    |
|                | V <sub>REFPF</sub> to AGND                 | 3                       |     | 15  | V    |
|                | V <sub>REFNF</sub> to AGND                 | –15                     |     | 0   | V    |
|                | V <sub>REFPF</sub> to V <sub>REFNF</sub>   | 3                       |     | 30  | V    |
| T <sub>A</sub> | Operating temperature                      | –40                     |     | 125 | °C   |

### 7.4 Thermal Information Package

| THERMAL METRIC <sup>(1)</sup> |                                              | DAC11001A, DAC91001,<br>DAC81001 | UNIT |
|-------------------------------|----------------------------------------------|----------------------------------|------|
|                               |                                              | PFB (TQFP)                       |      |
|                               |                                              | 48 PINS                          |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 51.0                             | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 10.3                             | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 16.2                             | °C/W |
| Ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 0.3                              | °C/W |
| Ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 16.0                             | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A                              | °C/W |

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics application report](#).

## 7.5 Electrical Characteristics

at  $T_A = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ ,  $V_{CC} = +15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5.5\text{ V}$ ,  $DV_{DD} = 3.3\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , see note<sup>(1)</sup> for  $V_{REFPF}$  and  $V_{REFNF}$ , 20-bit orderable used, OUT pin buffered with unity gain OPA827, ROFS, RCM, RFB unconnected, and all typical specifications at  $T_A = 25^{\circ}\text{C}$ , (unless otherwise noted)

| PARAMETER                 |                                                  | TEST CONDITIONS                                                                                                      | MIN  | TYP        | MAX | UNIT                        |
|---------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------|------------|-----|-----------------------------|
| <b>STATIC PERFORMANCE</b> |                                                  |                                                                                                                      |      |            |     |                             |
|                           | Resolution                                       | DAC11001A                                                                                                            | 20   |            |     | Bits                        |
|                           |                                                  | DAC91001                                                                                                             | 18   |            |     |                             |
|                           |                                                  | DAC81001                                                                                                             | 16   |            |     |                             |
| INL                       | Relative accuracy <sup>(2)</sup>                 | DAC11001A                                                                                                            | -4   |            | 4   | LSB                         |
|                           |                                                  | DAC11001A <sup>(3)(4)</sup>                                                                                          | -2.6 |            | 2.6 |                             |
|                           |                                                  | DAC11001A, $T_A = 25^{\circ}\text{C}$ <sup>(4)</sup>                                                                 | -2   |            | 2   |                             |
|                           |                                                  | DAC91001                                                                                                             | -1   |            | 1   |                             |
|                           |                                                  | DAC81001                                                                                                             | -1   |            | 1   |                             |
|                           | Relative accuracy drift over time <sup>(2)</sup> | $T_A = 25^{\circ}\text{C}$ , 1000 hrs                                                                                |      | $\pm 0.1$  |     | LSB                         |
| DNL                       | Differential nonlinearity <sup>(2)(3)</sup>      |                                                                                                                      | -1   |            | 1   | LSB                         |
|                           | Zero code error <sup>(4)</sup>                   | DAC11001A, $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only                | -4   |            | 4   | LSB                         |
|                           |                                                  | DAC11001A, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only            | -4   |            | 4   |                             |
|                           |                                                  | DAC11001A, $T_A = 25^{\circ}\text{C}$ , unipolar ranges only                                                         |      | $\pm 2$    |     |                             |
|                           |                                                  | DAC91001, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only             | -4   |            | 4   |                             |
|                           |                                                  | DAC81001, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only             | -4   |            | 4   |                             |
|                           | Zero code error temperature coefficient          | $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only                           |      | $\pm 0.04$ |     | ppm FSR/ $^{\circ}\text{C}$ |
|                           |                                                  | $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$ , code 0d into DAC, unipolar ranges only                       |      | $\pm 0.04$ |     |                             |
|                           | Gain error <sup>(2)(4)</sup>                     | DAC11001A, $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$                                                         | -8   |            | 8   | ppm of FSR                  |
|                           |                                                  | DAC11001A, $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$ , $V_{REFPF} = 3\text{ V}$ , $V_{REFNF} = -10\text{ V}$ | -8   |            | 8   |                             |
|                           |                                                  | DAC11001A, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$                                                     | -10  |            | 10  |                             |
|                           |                                                  | DAC11001A, $T_A = 25^{\circ}\text{C}$                                                                                |      | $\pm 2$    |     |                             |
|                           |                                                  | DAC91001, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$                                                      | -10  |            | 10  |                             |
|                           |                                                  | DAC81001, $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$                                                      | -10  |            | 10  |                             |
|                           | Gain error temperature coefficient               | $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$                                                                    |      | $\pm 0.04$ |     | ppm FSR/ $^{\circ}\text{C}$ |
|                           |                                                  | $T_A = 0^{\circ}\text{C}$ to $70^{\circ}\text{C}$ , $V_{REFPF} = 3\text{ V}$ , $V_{REFNF} = -10\text{ V}$            |      | $\pm 0.04$ |     |                             |
|                           |                                                  | $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$                                                                |      | $\pm 0.04$ |     |                             |

(1) Specified for the following pairs:  $V_{REFPF} = 5\text{ V}$  and  $V_{REFNF} = 0\text{ V}$ ;  $V_{REFPF} = 10\text{ V}$  and  $V_{REFNF} = 0\text{ V}$ ;  $V_{REFPF} = +5\text{ V}$  and  $V_{REFNF} = -5\text{ V}$ ;  $V_{REFPF} = +10\text{ V}$  and  $V_{REFNF} = -10\text{ V}$ .

(2) Calculated between code 0d to 1048575d for DAC11001A, code 0d to 262143d for DAC91001, code 0d to 65535d for DAC81001.

(3) With device temperature calibration mode enabled and used.

(4) Specified by design, not production tested.

## Electrical Characteristics (continued)

at  $T_A = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ ,  $V_{CC} = +15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5.5\text{ V}$ ,  $DV_{DD} = 3.3\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , see note<sup>(1)</sup> for  $V_{REFPF}$  and  $V_{REFNF}$ , 20-bit orderable used, OUT pin buffered with unity gain OPA827, ROFS, RCM, RFB unconnected, and all typical specifications at  $T_A = 25^{\circ}\text{C}$ , (unless otherwise noted)

| PARAMETER               |                                          | TEST CONDITIONS                                                                                                       | MIN | TYP   | MAX | UNIT          |
|-------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----|-------|-----|---------------|
|                         | Positive full-scale error <sup>(4)</sup> | DAC11001A, T <sub>A</sub> = 0°C to 70°C, code 1048575d into DAC                                                       | −8  |       | 8   | LSB           |
|                         |                                          | DAC11001A, T <sub>A</sub> = 0°C to 70°C, code 1048575d into DAC, V <sub>REFPF</sub> = 3 V, V <sub>REFNF</sub> = −10 V | −6  |       | 6   |               |
|                         |                                          | DAC11001A, T <sub>A</sub> = −40°C to +125°C, code 1048575d into DAC                                                   | −10 |       | 10  |               |
|                         |                                          | DAC11001A, T <sub>A</sub> = 25°C, code 1048575d into DAC                                                              |     | ±2    |     |               |
|                         |                                          | DAC91001, T <sub>A</sub> = −40°C to +125°C, code 262143d into DAC                                                     | −10 |       | 10  |               |
|                         |                                          | DAC81001, T <sub>A</sub> = −40°C to +125°C, code 65535d into DAC                                                      | −10 |       | 10  |               |
|                         | Full-scale error temperature coefficient | T <sub>A</sub> = 0°C to 70°C                                                                                          |     | ±0.04 |     | ppm<br>FSR/°C |
|                         |                                          | T <sub>A</sub> = 0°C to 70°C, V <sub>REFPF</sub> = 3 V, V <sub>REFNF</sub> = −10 V                                    |     | ±0.04 |     |               |
|                         |                                          | T <sub>A</sub> = −40°C to +125°C                                                                                      |     | ±0.04 |     |               |
| OUTPUT CHARACTERISTICS  |                                          |                                                                                                                       |     |       |     |               |
|                         | Headroom                                 | From V <sub>REFPF</sub> to V <sub>CC</sub>                                                                            | 3   |       |     | V             |
|                         | Footroom                                 | From V <sub>REFNF</sub> to V <sub>SS</sub>                                                                            | 3   |       |     | V             |
|                         | DC impedance                             | From ROFS to RCM                                                                                                      |     | 5     |     | kΩ            |
|                         |                                          | From RCM to RFB                                                                                                       |     | 5     |     |               |
| Z <sub>O</sub>          | DC output impedance                      |                                                                                                                       |     | 2.5   |     | kΩ            |
|                         | Power supply rejection ratio (dc)        | T <sub>A</sub> = 25°C, V <sub>CC</sub> = 15 V ± 20%, V <sub>SS</sub> = −15 V                                          |     | 1.5   |     | μV/V          |
|                         |                                          | T <sub>A</sub> = 25°C, V <sub>CC</sub> = 15 V, V <sub>SS</sub> = −15 V ± 20%                                          |     | 1     |     |               |
|                         | Output voltage drift over time           | T <sub>A</sub> = 25°C, V <sub>OUT</sub> = midscale, 1000 hr                                                           |     | 1     |     | ppm of<br>FSR |
| VOLTAGE REFERENCE INPUT |                                          |                                                                                                                       |     |       |     |               |
|                         | Reference input impedance (REFPF)        | DAC at midscale, V <sub>REFPF</sub> = 10 V, V <sub>REFNF</sub> = 0 V                                                  |     | 5.5   |     | kΩ            |
|                         | Reference input impedance (REFNF)        | DAC at midscale, V <sub>REFPF</sub> = 10 V, V <sub>REFNF</sub> = 0 V                                                  |     | 7     |     |               |

## Electrical Characteristics (continued)

at  $T_A = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ ,  $V_{CC} = +15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5.5\text{ V}$ ,  $DV_{DD} = 3.3\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , see note<sup>(1)</sup> for  $V_{REFPF}$  and  $V_{REFNF}$ , 20-bit orderable used, OUT pin buffered with unity gain OPA827, ROFS, RCM, RFB unconnected, and all typical specifications at  $T_A = 25^{\circ}\text{C}$ , (unless otherwise noted)

| PARAMETER                  | TEST CONDITIONS                             | MIN                                                                                                                                                           | TYP  | MAX | UNIT                         |
|----------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------------------------------|
| <b>DYNAMIC PERFORMANCE</b> |                                             |                                                                                                                                                               |      |     |                              |
| $t_s$                      | Output voltage settling time <sup>(5)</sup> | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , full-scale settling to 0.1%FSR                                                                         | 1    |     | $\mu\text{s}$                |
|                            |                                             | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , full-scale settling to $\pm 1\text{ LSB}$                                                              | 2.5  |     |                              |
|                            |                                             | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , 1-mV step settling to $\pm 1\text{ LSB}$                                                               | 2.5  |     |                              |
| SR                         | Slew rate                                   | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , full-scale step, measured at OUT pin                                                                   | 50   |     | $\text{V}/\mu\text{s}$       |
|                            | Power-on glitch magnitude                   | Measured at unbuffered DAC voltage output, $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$                                                               | -0.2 |     | V                            |
| $V_n$                      | Output noise                                | 0.1-Hz to 10-Hz, DAC at midscale, $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$                                                                        | 0.4  |     | $\mu\text{Vpp}$              |
|                            |                                             | 100-kHz bandwidth, DAC at midscale, $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$                                                                      | 3    |     | $\mu\text{Vrms}$             |
|                            | Output noise density                        | Measured at 1 kHz, 10 kHz, 100 kHz, DAC at mid scale, $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$                                                    | 7    |     | $\text{nV}/\sqrt{\text{Hz}}$ |
| SFDR                       | Spurious free dynamic range                 | DAC update rate = 400 kHz, $f_{OUT} = 1\text{ kHz}$ , $V_{OUTPP} = 0\text{ V}$ to $10\text{ V}$                                                               | -105 |     | dB                           |
|                            |                                             | DAC update rate = 400 kHz, $f_{OUT} = 1\text{ kHz}$ , $V_{OUTPP} = 3\text{ V}$ to $-10\text{ V}$                                                              | -105 |     | dB                           |
| THD                        | Total harmonic distortion                   | DAC update rate = 400 kHz, $f_{OUT} = 1\text{ kHz}$ , $V_{OUTPP} = 0\text{ V}$ to $10\text{ V}$                                                               | -105 |     | dB                           |
|                            |                                             | DAC update rate = 400 kHz, $f_{OUT} = 1\text{ kHz}$ , $V_{OUTPP} = 3\text{ V}$ to $-10\text{ V}$                                                              | -105 |     | dB                           |
|                            | Power supply rejection ratio (ac)           | 200-mV 50-Hz or 60-Hz sine wave superimposed on $V_{SS}$ , $V_{CC} = 15\text{ V}$                                                                             | 95   |     | dB                           |
|                            |                                             | 200-mV 50 Hz or 60 Hz sine wave superimposed on $V_{CC}$ , $V_{SS} = -15\text{ V}$                                                                            | 95   |     | dB                           |
|                            | Code change glitch impulse                  | $\pm 1\text{ LSB}$ change around mid code (including feedthrough), $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , measured at output of buffer op amp | 1    |     | nV-s                         |
|                            | Code change glitch impulse magnitude        | $\pm 1\text{ LSB}$ change around mid code (including feedthrough), $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , measured at output of buffer op amp | 5    |     | mV                           |
|                            | Reference feedthrough                       | $V_{REFPF} = 10\text{ V} \pm 10\%$ , $V_{REFNF} = 0\text{ V}$ , frequency = 100 Hz, DAC at zero scale                                                         | -90  |     | dB                           |
|                            | Reference feedthrough                       | $V_{REFNF} = -10\text{ V} \pm 10\%$ , $V_{REFPF} = 10\text{ V}$ , frequency = 100 Hz, DAC at full scale                                                       | -90  |     | dB                           |
|                            | Digital feedthrough                         | At SCLK = 1 MHz, DAC output static at midscale, 10-V range                                                                                                    | 1    |     | nV-s                         |

(5) Adaptive TnH mode. TnH action is disabled for large code steps. For small steps, TnH action happens with a hold time of 1.2 $\mu\text{s}$ .

## Electrical Characteristics (continued)

at  $T_A = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ ,  $V_{CC} = +15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5.5\text{ V}$ ,  $DV_{DD} = 3.3\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , see note<sup>(1)</sup> for  $V_{REFPF}$  and  $V_{REFNF}$ , 20-bit orderable used, OUT pin buffered with unity gain OPA827, ROFS, RCM, RFB unconnected, and all typical specifications at  $T_A = 25^{\circ}\text{C}$ , (unless otherwise noted)

| PARAMETER              |                                         | TEST CONDITIONS                                                                                                   | MIN              | TYP     | MAX | UNIT          |
|------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|---------|-----|---------------|
| <b>DIGITAL INPUTS</b>  |                                         |                                                                                                                   |                  |         |     |               |
|                        | Hysteresis voltage                      |                                                                                                                   |                  | 0.4     |     | V             |
|                        | Input current                           |                                                                                                                   |                  | $\pm 5$ |     | $\mu\text{A}$ |
|                        | Pin capacitance                         | Per pin                                                                                                           |                  | 10      |     | pF            |
| <b>DIGITAL OUTPUTS</b> |                                         |                                                                                                                   |                  |         |     |               |
| $V_{OL}$               | Output low voltage                      | sinking 200 $\mu\text{A}$                                                                                         |                  |         | 0.4 | V             |
| $V_{OH}$               | Output high voltage                     | sourcing 200 $\mu\text{A}$                                                                                        | $IOV_{DD} - 0.5$ |         |     | V             |
|                        | High impedance leakage                  |                                                                                                                   |                  | $\pm 5$ |     | $\mu\text{A}$ |
|                        | High impedance output capacitance       |                                                                                                                   |                  | 10      |     | pF            |
| <b>POWER</b>           |                                         |                                                                                                                   |                  |         |     |               |
| $I_{AVDD}$             | Current flowing into $AV_{DD}$          | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  |         | 1.5 | mA            |
| $I_{VCC}$              | Current flowing into $V_{CC}$           | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  |         | 7   | mA            |
| $I_{VSS}$              | Current flowing into $V_{SS}$           | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  |         | 7   | mA            |
| $I_{DVDD}$             | Current flowing into $DV_{DD}$          | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  | 0.5     |     | mA            |
| $I_{IOVDD}$            | Current flowing into $IOV_{DD}$         | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code, all digital input pins static at $IOV_{DD}$ |                  | 0.1     |     | mA            |
| $I_{REFPF}$            | Reference input current ( $V_{REFPF}$ ) | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  |         | 5   | mA            |
| $I_{REFNF}$            | Reference input current ( $V_{REFNF}$ ) | $V_{REFPF} = 10\text{ V}$ , $V_{REFNF} = 0\text{ V}$ , midscale code                                              |                  |         | 5   | mA            |

## 7.6 Timing Requirements: Write, $4.5\text{ V} \leq \text{DV}_{\text{DD}} \leq 5.5\text{ V}$

all input signals are specified with  $t_R = t_F = 1\text{ ns/V}$  (10% to 90% of  $\text{IOV}_{\text{DD}}$ ) and timed from a voltage level of  $(V_{\text{IL}} + V_{\text{IH}}) / 2$ , SDO loaded with 20 pF, and  $T_A = -40^\circ\text{C}$  to  $+125^\circ\text{C}$  (unless otherwise noted)

|                       |                                                                                                                                                                 | MIN | NOM | MAX | UNIT |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{\text{SCLK}}$     | SCLK frequency, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                       |     |     | 33  | MHz  |
|                       | SCLK frequency, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                    |     |     | 50  |      |
| $t_{\text{SCLKHIGH}}$ | SCLK high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                       | 15  |     |     | ns   |
|                       | SCLK high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                    | 10  |     |     |      |
| $t_{\text{SCLKLOW}}$  | SCLK low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                        | 15  |     |     | ns   |
|                       | SCLK low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                     | 10  |     |     |      |
| $t_{\text{SDIS}}$     | SDI setup, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                            | 13  |     |     | ns   |
|                       | SDI setup, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                         | 8   |     |     |      |
| $t_{\text{SDIH}}$     | SDI hold, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                             | 13  |     |     | ns   |
|                       | SDI hold, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                          | 8   |     |     |      |
| $t_{\text{CSS}}$      | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                           | 23  |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                        | 18  |     |     |      |
| $t_{\text{CSH}}$      | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                            | 15  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                         | 10  |     |     |      |
| $t_{\text{CSHIGH}}$   | $\overline{\text{SYNC}}$ high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                   | 55  |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                | 50  |     |     |      |
| $t_{\text{CSIGNORE}}$ | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                 | 10  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                              | 5   |     |     |      |
| $t_{\text{LDACSL}}$   | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$    | 50  |     |     | ns   |
|                       | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ | 50  |     |     |      |
| $t_{\text{LDACW}}$    | $\overline{\text{LDAC}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                    | 20  |     |     | ns   |
|                       | $\overline{\text{LDAC}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                 | 20  |     |     |      |
| $t_{\text{CLR W}}$    | $\overline{\text{CLR}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                     | 20  |     |     | ns   |
|                       | $\overline{\text{CLR}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                  | 20  |     |     |      |

## 7.7 Timing Requirements: Write, $2.7\text{ V} \leq \text{DV}_{\text{DD}} < 4.5\text{ V}$

all input signals are specified with  $t_R = t_F = 1\text{ ns/V}$  (10% to 90% of  $\text{IOV}_{\text{DD}}$ ) and timed from a voltage level of  $(V_{\text{IL}} + V_{\text{IH}}) / 2$ , SDO loaded with 20 pF, and  $T_A = -40^\circ\text{C}$  to  $+125^\circ\text{C}$  (unless otherwise noted)

|                       |                                                                                                                                                                 | MIN | NOM | MAX | UNIT |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{\text{SCLK}}$     | SCLK frequency, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                       |     |     | 20  | MHz  |
|                       | SCLK frequency, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                    |     |     | 25  |      |
| $t_{\text{SCLKHIGH}}$ | SCLK high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                       | 25  |     |     | ns   |
|                       | SCLK high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                    | 20  |     |     |      |
| $t_{\text{SCLKLOW}}$  | SCLK low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                        | 25  |     |     | ns   |
|                       | SCLK low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                     | 20  |     |     |      |
| $t_{\text{SDIS}}$     | SDI setup, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                            | 21  |     |     | ns   |
|                       | SDI setup, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                         | 16  |     |     |      |
| $t_{\text{SDIH}}$     | SDI hold, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                             | 21  |     |     | ns   |
|                       | SDI hold, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                          | 16  |     |     |      |
| $t_{\text{CSS}}$      | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                           | 41  |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                        | 36  |     |     |      |
| $t_{\text{CSH}}$      | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                            | 25  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                         | 20  |     |     |      |
| $t_{\text{CSHIGH}}$   | $\overline{\text{SYNC}}$ high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                   | 100 |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                | 100 |     |     |      |
| $t_{\text{CSIGNORE}}$ | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                 | 10  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                              | 5   |     |     |      |
| $t_{\text{LDACSL}}$   | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$    | 100 |     |     | ns   |
|                       | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ | 100 |     |     |      |
| $t_{\text{LDACW}}$    | $\overline{\text{LDAC}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                    | 40  |     |     | ns   |
|                       | $\overline{\text{LDAC}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                 | 40  |     |     |      |
| $t_{\text{CLR W}}$    | $\overline{\text{CLR}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                     | 40  |     |     | ns   |
|                       | $\overline{\text{CLR}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                  | 40  |     |     |      |

## 7.8 Timing Requirements: Read and Daisy-Chain Write, $4.5\text{ V} \leq \text{DV}_{\text{DD}} \leq 5.5\text{ V}$

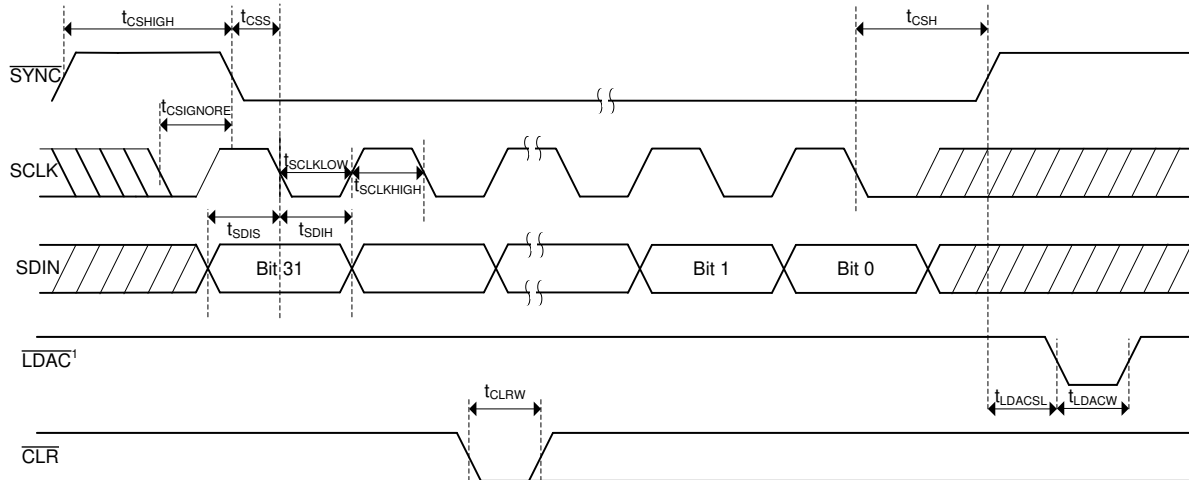
all input signals are specified with  $t_{\text{R}} = t_{\text{F}} = 1\text{ ns/V}$  (10% to 90% of  $\text{IOV}_{\text{DD}}$ ) and timed from a voltage level of  $(V_{\text{IL}} + V_{\text{IH}}) / 2$ , SDO loaded with 20 pF, and  $T_{\text{A}} = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$  (unless otherwise noted)

|                       |                                                                                                                                                                 |                                                                                  | MIN | NOM | MAX | UNIT |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{\text{SCLK}}$     | SCLK frequency                                                                                                                                                  | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$    |     |     | 10  | MHz  |
|                       |                                                                                                                                                                 | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$    |     |     | 20  |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$ |     |     | 15  |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$ |     |     | 30  |      |
| $t_{\text{SCLKHIGH}}$ | SCLK high time                                                                                                                                                  | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$    | 50  |     |     | ns   |
|                       |                                                                                                                                                                 | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$    | 25  |     |     |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$ | 33  |     |     |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$ | 16  |     |     |      |
| $t_{\text{SCLKLOW}}$  | SCLK low time                                                                                                                                                   | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$    | 50  |     |     | ns   |
|                       |                                                                                                                                                                 | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$    | 25  |     |     |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$ | 33  |     |     |      |
|                       |                                                                                                                                                                 | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$ | 16  |     |     |      |
| $t_{\text{SDIS}}$     | SDI setup, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                            |                                                                                  | 13  |     |     | ns   |
|                       | SDI setup, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                         |                                                                                  | 8   |     |     |      |
| $t_{\text{SDIH}}$     | SDI hold, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                             |                                                                                  | 13  |     |     | ns   |
|                       | SDI hold, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                          |                                                                                  | 8   |     |     |      |
| $t_{\text{CSS}}$      | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                           |                                                                                  | 30  |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                        |                                                                                  | 20  |     |     |      |
| $t_{\text{CSH}}$      | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                            |                                                                                  | 15  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                         |                                                                                  | 10  |     |     |      |
| $t_{\text{CSHIGH}}$   | $\overline{\text{SYNC}}$ high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                   |                                                                                  | 55  |     |     | ns   |
|                       | $\overline{\text{SYNC}}$ high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                |                                                                                  | 50  |     |     |      |
| $t_{\text{CSIGNORE}}$ | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                 |                                                                                  | 10  |     |     | ns   |
|                       | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                              |                                                                                  | 5   |     |     |      |
| $t_{\text{LDACSL}}$   | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$    |                                                                                  | 50  |     |     | ns   |
|                       | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ |                                                                                  | 50  |     |     |      |
| $t_{\text{LDACW}}$    | $\overline{\text{LDAC}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                    |                                                                                  | 20  |     |     | ns   |
|                       | $\overline{\text{LDAC}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                 |                                                                                  | 20  |     |     |      |
| $t_{\text{CLRW}}$     | $\overline{\text{CLR}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                     |                                                                                  | 20  |     |     | ns   |
|                       | $\overline{\text{CLR}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                  |                                                                                  | 20  |     |     |      |
| $t_{\text{SDODLY}}$   | SCLK rising edge to SDO valid data, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$                                               |                                                                                  | 0   |     | 35  | ns   |
|                       | SCLK rising edge to SDO valid data, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$                                            |                                                                                  | 0   |     | 25  |      |
|                       | SCLK falling edge to SDO valid data, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$                                              |                                                                                  | 0   |     | 35  |      |
|                       | SCLK falling edge to SDO valid data, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$                                           |                                                                                  | 0   |     | 25  |      |
| $t_{\text{SDOZ}}$     | $\overline{\text{SYNC}}$ rising edge to SDO HiZ, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                      |                                                                                  | 0   |     | 20  | ns   |
|                       | $\overline{\text{SYNC}}$ rising edge to SDO HiZ, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                   |                                                                                  | 0   |     | 20  |      |

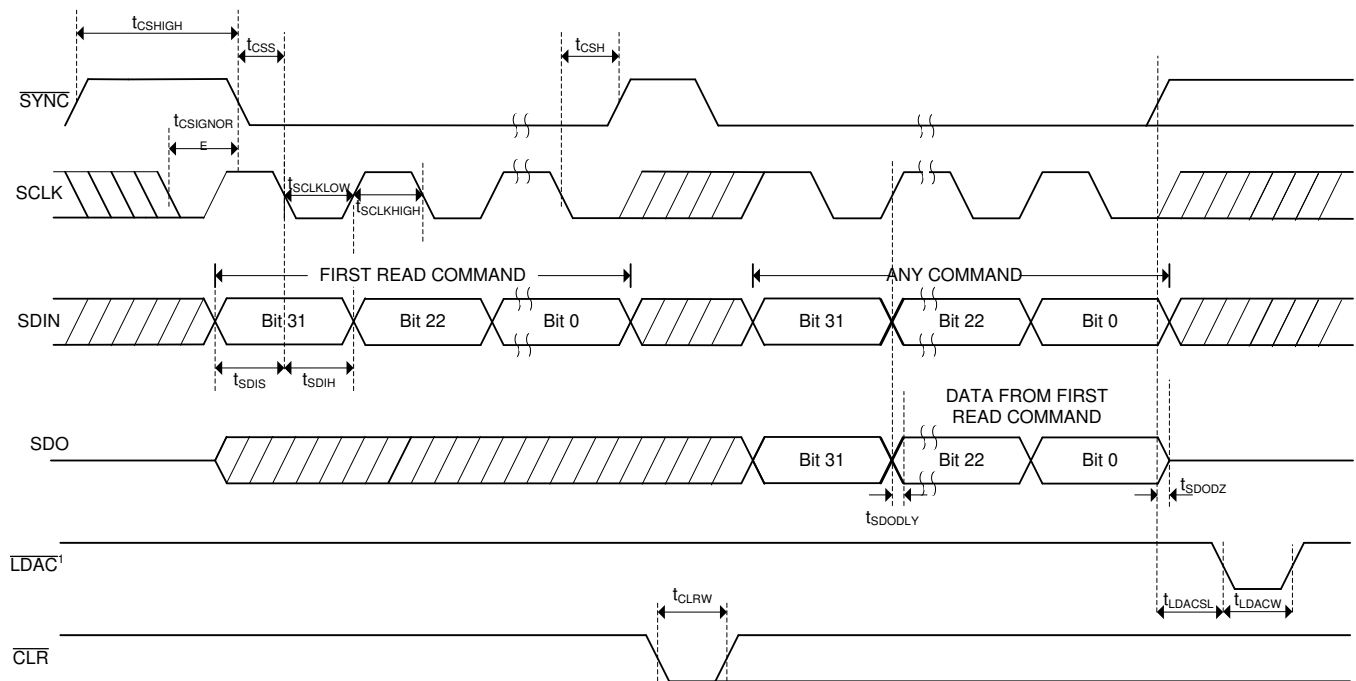
## 7.9 Timing Requirements: Read and Daisy-Chain Write, $2.7\text{ V} \leq \text{DV}_{\text{DD}} < 4.5\text{ V}$

all input signals are specified with  $t_{\text{R}} = t_{\text{F}} = 1\text{ ns/V}$  (10% to 90% of  $\text{IOV}_{\text{DD}}$ ) and timed from a voltage level of  $(V_{\text{IL}} + V_{\text{IH}}) / 2$ , SDO loaded with 20 pF, and  $T_{\text{A}} = -40^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$  (unless otherwise noted)

|                       |                                                                                                                                                              |                                                                                                                                                                 | MIN | NOM | MAX | UNIT |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| $f_{\text{SCLK}}$     | SCLK frequency                                                                                                                                               | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$                                                                                   |     |     | 8   | MHz  |
|                       |                                                                                                                                                              | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$                                                                                   |     |     | 16  |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$                                                                                |     |     | 10  |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$                                                                                |     |     | 20  |      |
| $t_{\text{SCLKHIGH}}$ | SCLK high time                                                                                                                                               | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$                                                                                   | 62  |     |     | ns   |
|                       |                                                                                                                                                              | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$                                                                                   | 31  |     |     |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$                                                                                | 50  |     |     |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$                                                                                | 25  |     |     |      |
| $t_{\text{SCLKLOW}}$  | SCLK low time                                                                                                                                                | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$                                                                                   | 62  |     |     | ns   |
|                       |                                                                                                                                                              | $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$                                                                                   | 31  |     |     |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$                                                                                | 50  |     |     |      |
|                       |                                                                                                                                                              | $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$                                                                                | 25  |     |     |      |
| $t_{\text{SDIS}}$     | SDI setup, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                         |                                                                                                                                                                 | 21  |     |     | ns   |
|                       |                                                                                                                                                              | SDI setup, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                         | 16  |     |     |      |
| $t_{\text{SDIH}}$     | SDI hold, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                                          |                                                                                                                                                                 | 21  |     |     | ns   |
|                       |                                                                                                                                                              | SDI hold, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                                          | 16  |     |     |      |
| $t_{\text{CSS}}$      | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                        |                                                                                                                                                                 | 41  |     |     | ns   |
|                       |                                                                                                                                                              | $\overline{\text{SYNC}}$ falling edge to SCLK falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                        | 36  |     |     |      |
| $t_{\text{CSH}}$      | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                         |                                                                                                                                                                 | 25  |     |     | ns   |
|                       |                                                                                                                                                              | SCLK falling edge to $\overline{\text{SYNC}}$ rising edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                         | 20  |     |     |      |
| $t_{\text{CSHIGH}}$   | $\overline{\text{SYNC}}$ high time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                |                                                                                                                                                                 | 100 |     |     | ns   |
|                       |                                                                                                                                                              | $\overline{\text{SYNC}}$ high time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                | 100 |     |     |      |
| $t_{\text{CSIGNORE}}$ | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                              |                                                                                                                                                                 | 10  |     |     | ns   |
|                       |                                                                                                                                                              | SCLK falling edge to $\overline{\text{SYNC}}$ ignore, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                              | 5   |     |     |      |
| $t_{\text{LDACSL}}$   | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ |                                                                                                                                                                 | 100 |     |     | ns   |
|                       |                                                                                                                                                              | Synchronous update: $\overline{\text{SYNC}}$ rising edge to $\overline{\text{LDAC}}$ falling edge, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ | 100 |     |     |      |
| $t_{\text{LDACW}}$    | $\overline{\text{LDAC}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                 |                                                                                                                                                                 | 40  |     |     | ns   |
|                       |                                                                                                                                                              | $\overline{\text{LDAC}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                 | 40  |     |     |      |
| $t_{\text{CLRW}}$     | $\overline{\text{CLR}}$ low time, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                                  |                                                                                                                                                                 | 40  |     |     | ns   |
|                       |                                                                                                                                                              | $\overline{\text{CLR}}$ low time, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                                  | 40  |     |     |      |
| $t_{\text{SDODLY}}$   | SCLK rising edge to SDO valid data, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 0$                                            |                                                                                                                                                                 | 0   |     | 40  | ns   |
|                       |                                                                                                                                                              | SCLK rising edge to SDO valid data, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 0$                                            | 0   |     | 30  |      |
|                       |                                                                                                                                                              | SCLK rising edge to SDO valid data, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$ , $\text{FSDO} = 1$                                               | 0   |     | 40  |      |
|                       |                                                                                                                                                              | SCLK rising edge to SDO valid data, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$ , $\text{FSDO} = 1$                                            | 0   |     | 30  |      |
| $t_{\text{SDOZ}}$     | $\overline{\text{SYNC}}$ rising edge to SDO HiZ, $1.7\text{ V} \leq \text{IOV}_{\text{DD}} < 2.7\text{ V}$                                                   |                                                                                                                                                                 | 0   |     | 20  | ns   |
|                       |                                                                                                                                                              | $\overline{\text{SYNC}}$ rising edge to SDO HiZ, $2.7\text{ V} \leq \text{IOV}_{\text{DD}} \leq 5.5\text{ V}$                                                   | 0   |     | 20  |      |



**Figure 1. Serial Interface Write Timing: Standalone Mode**



**Figure 2. Serial Interface Read and Write Timing: Daisy-Chain Mode**

## 7.10 Typical Characteristics

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)

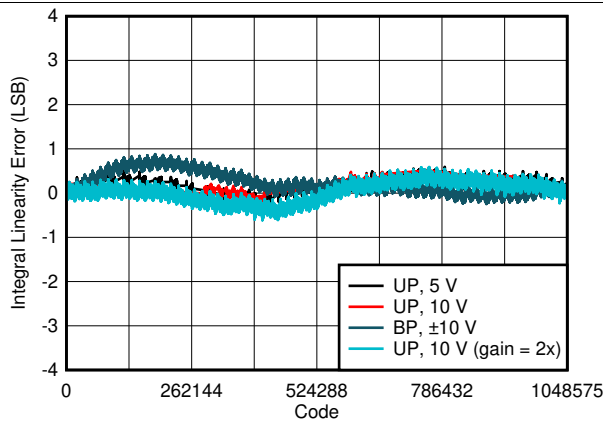


Figure 3. Integral Linearity Error vs Digital Input Code

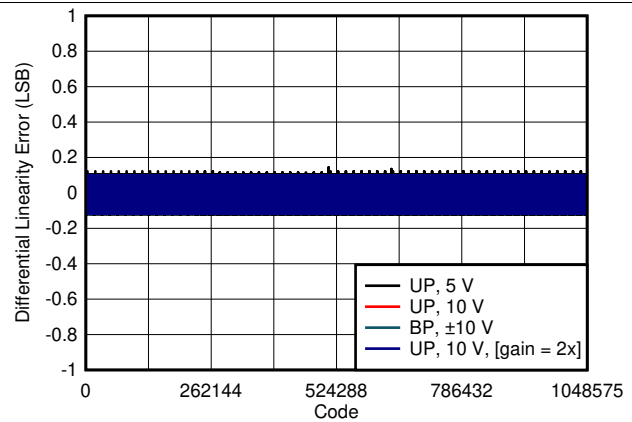


Figure 4. Differential Linearity Error vs Digital Input Code

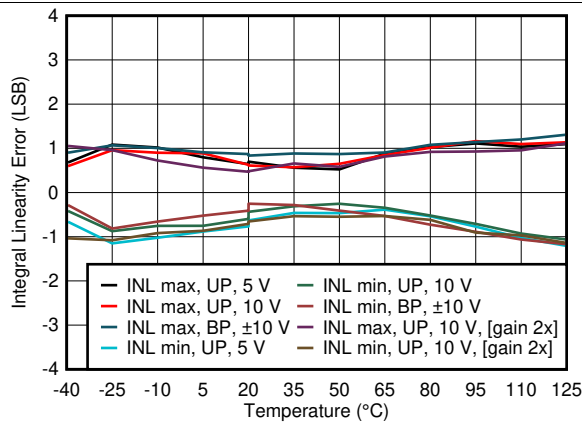
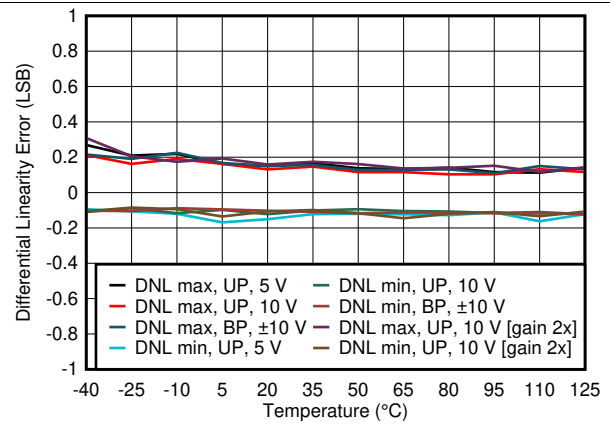
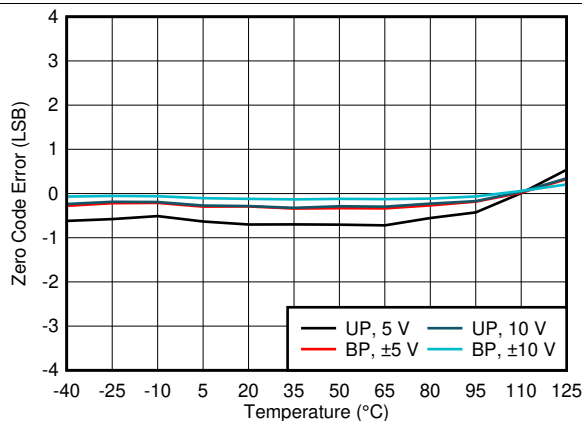


Figure 5. Integral Linearity Error vs Temperature



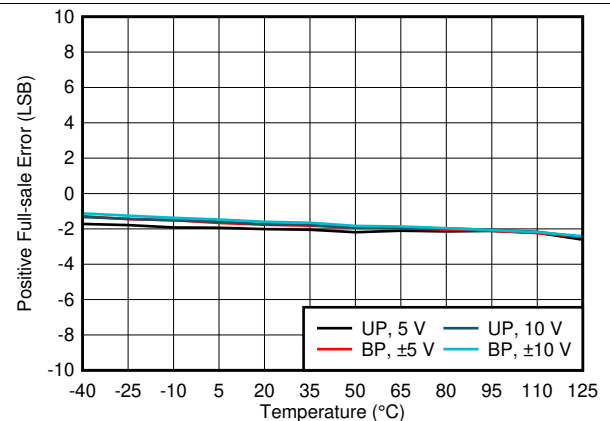
Temperature calibration enabled

Figure 6. Differential Linearity Error vs Temperature



Temperature calibration enabled

Figure 7. Zero Code Error vs Temperature

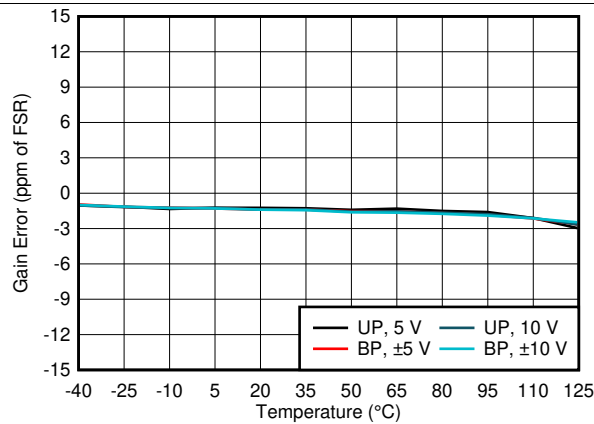


Temperature calibration enabled

Figure 8. Positive Full-Scale Error vs Temperature

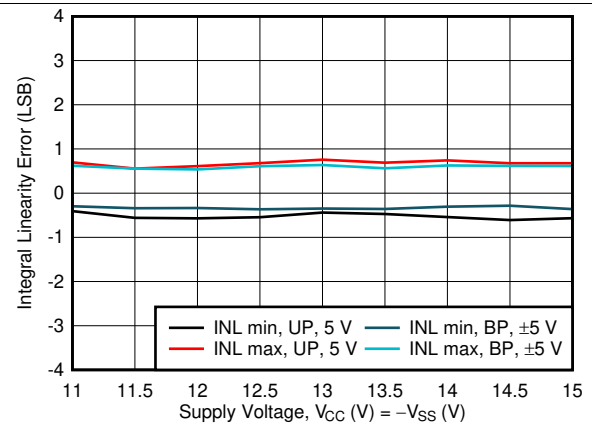
## Typical Characteristics (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)

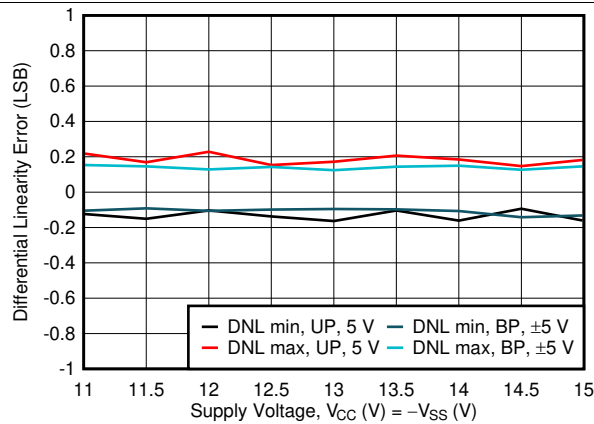


Temperature calibration enabled

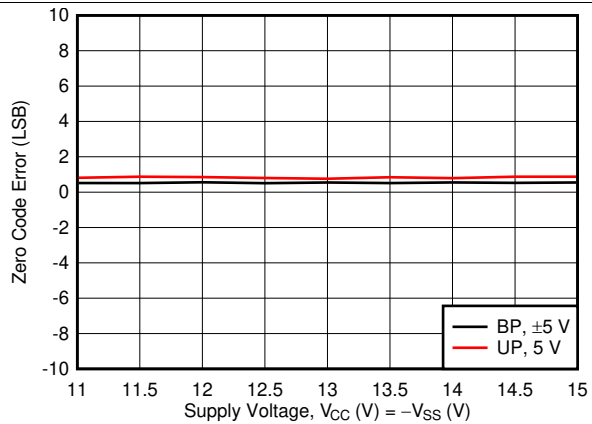
**Figure 9. Gain Error vs Temperature**



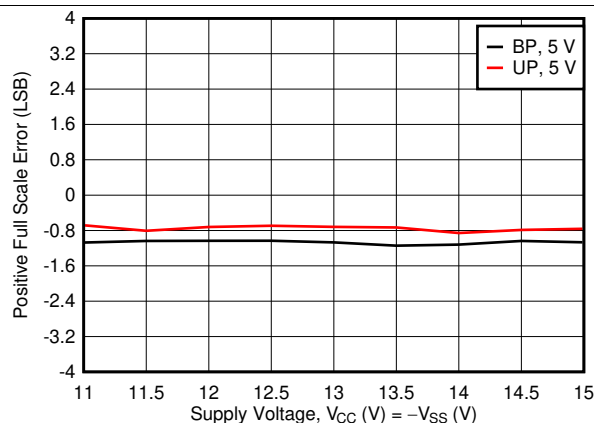
**Figure 10. Integral Linearity Error vs Supply Voltage**



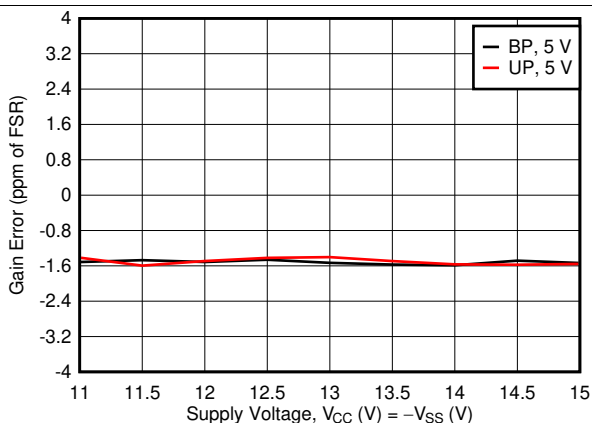
**Figure 11. Differential Linearity Error vs Supply Voltage**



**Figure 12. Zero Code Error vs Supply Voltage**



**Figure 13. Positive Full-Scale Error vs Supply Voltage**



**Figure 14. Gain Error vs Supply Voltage**

## Typical Characteristics (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)

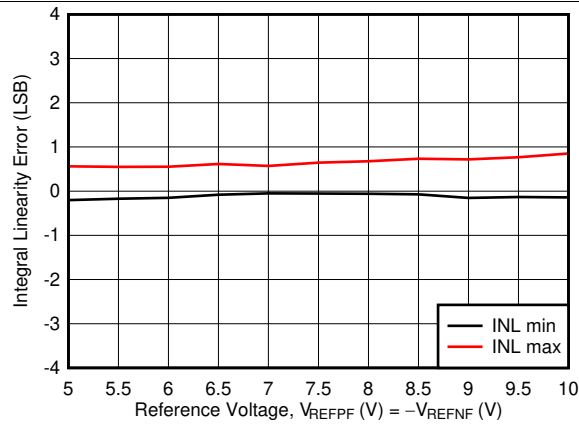


Figure 15. Integral Linearity Error vs Reference Voltage

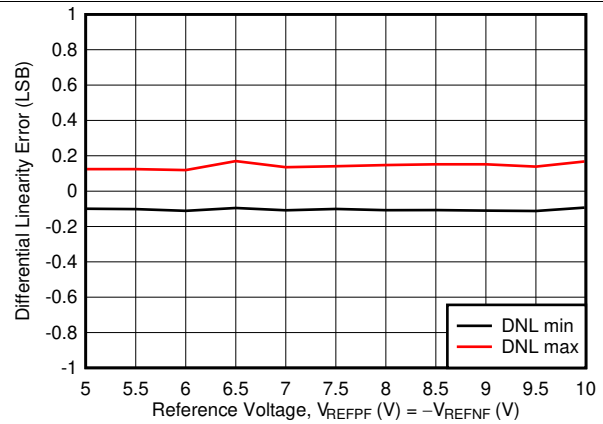


Figure 16. Differential Linearity Error vs Reference Voltage

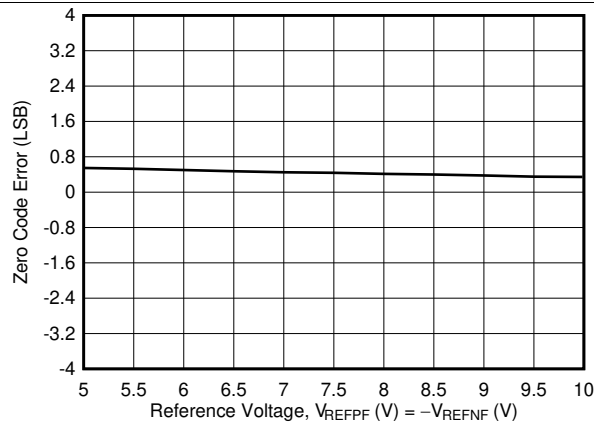


Figure 17. Zero Code Error vs Reference Voltage

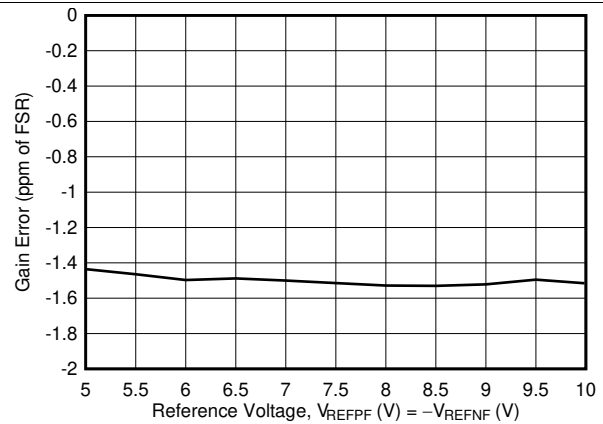


Figure 18. Gain Error vs Reference Voltage

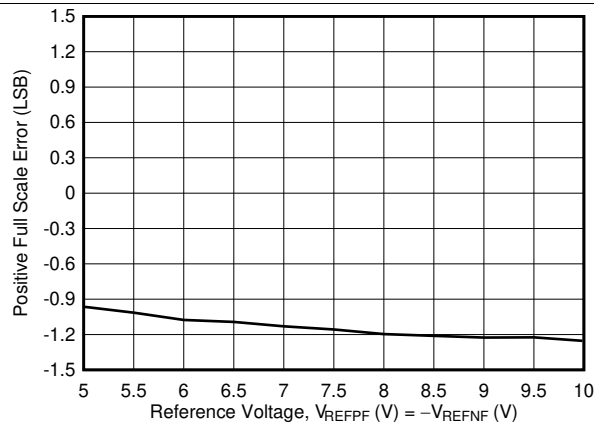


Figure 19. Positive Full-Scale Error vs Reference Voltage

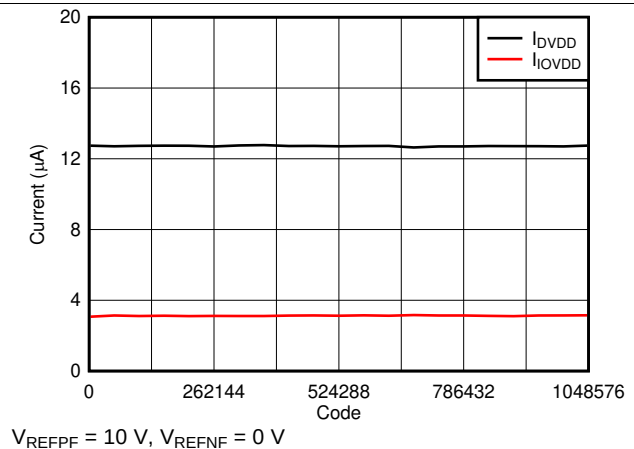
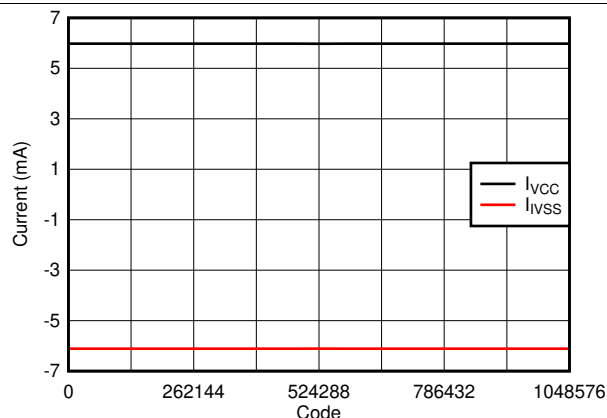


Figure 20. Supply Current ( $DV_{DD}$  and  $IOV_{DD}$ ) vs Digital Input Code

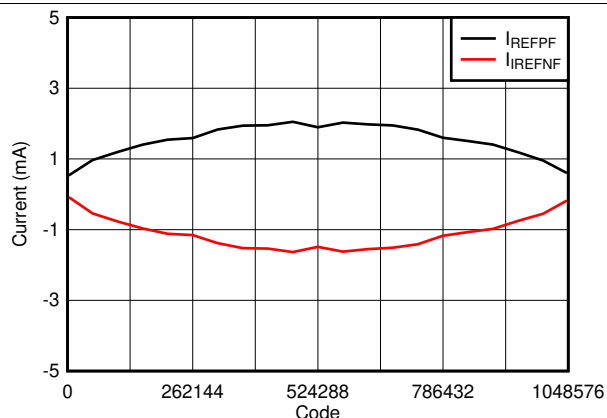
## Typical Characteristics (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)



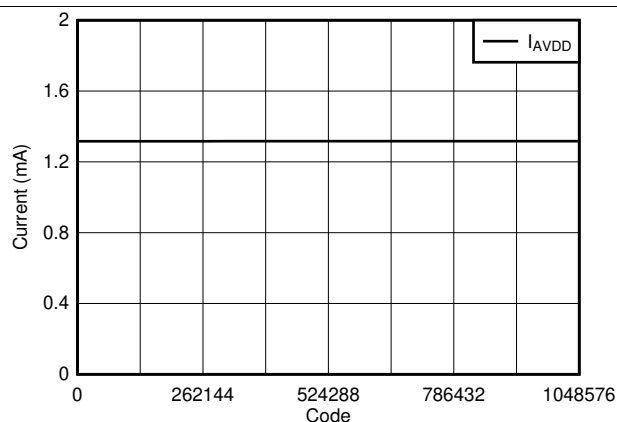
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 21. Supply Current ( $V_{CC}$  and  $V_{SS}$ ) vs Digital Input Code



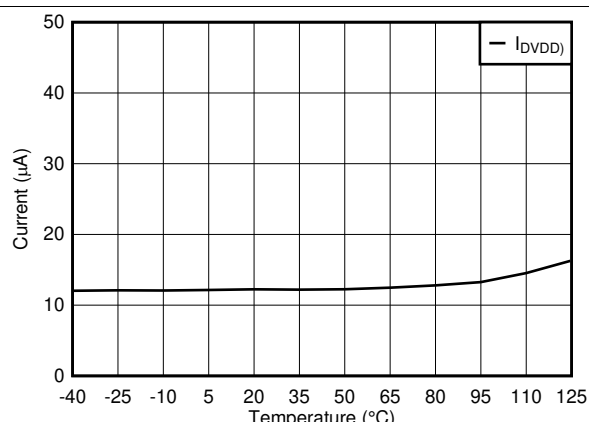
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 22. Reference Current ( $V_{REFPF}$  and  $V_{REFNF}$ ) vs Digital Input Code



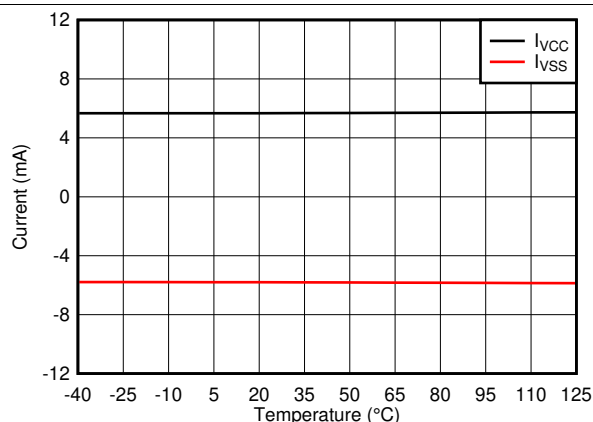
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 23. Supply Current ( $AV_{DD}$ ) vs Digital Input Code



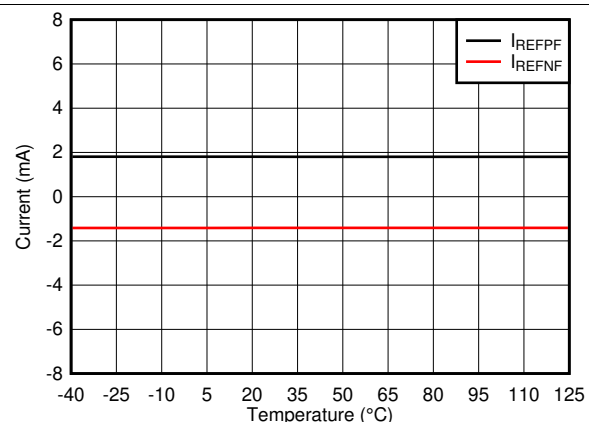
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC at midcode

Figure 24. Supply Current ( $DV_{DD}$ ) vs Temperature



$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC at midcode

Figure 25. Supply Current ( $V_{CC}$  and  $V_{SS}$ ) vs Temperature

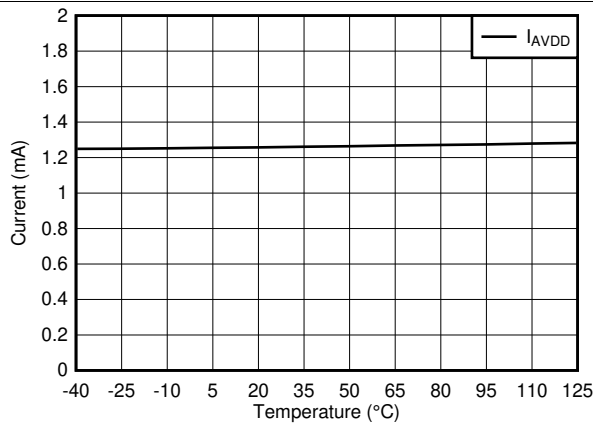


$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC at midcode

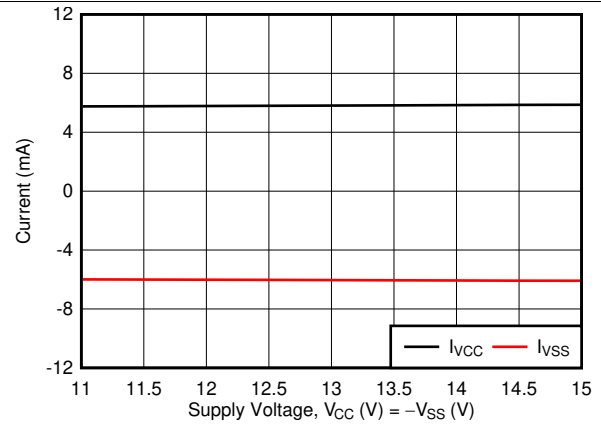
Figure 26. Reference Current ( $V_{REFPF}$  and  $V_{REFNF}$ ) vs Temperature

## Typical Characteristics (continued)

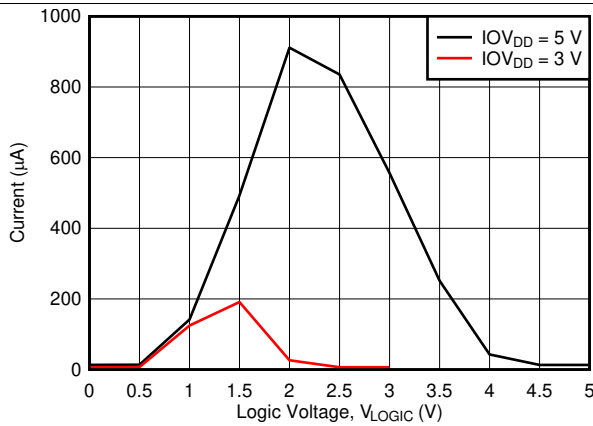
at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)



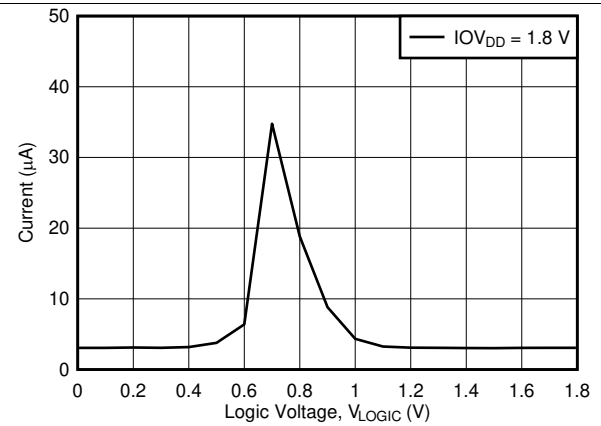
**Figure 27. Supply Current ( $AV_{DD}$ ) vs Temperature**



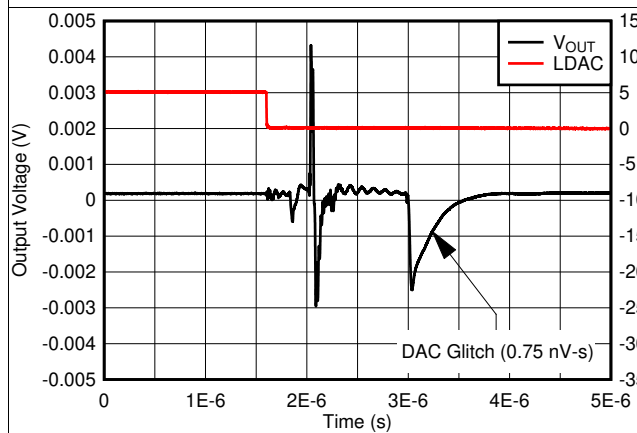
**Figure 28. Supply Current ( $V_{CC}$  and  $V_{SS}$ ) vs Supply Voltage**



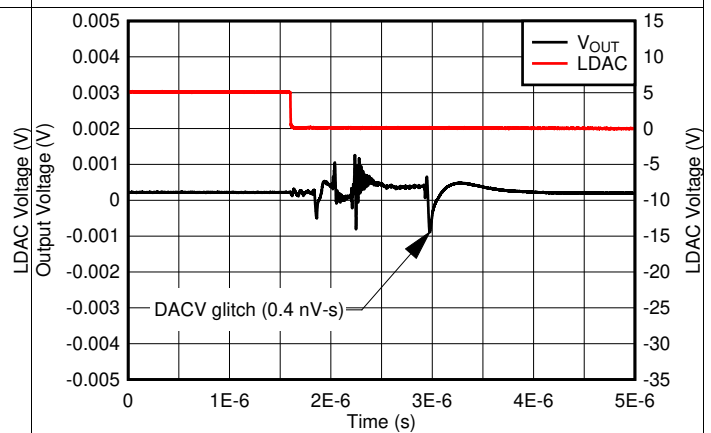
**Figure 29. Supply Current ( $IOV_{DD}$ ) vs Input Pin Logic Level**



**Figure 30. Supply Current ( $IOV_{DD} = 1.8\text{ V}$ ) vs Input Pin Logic Level**



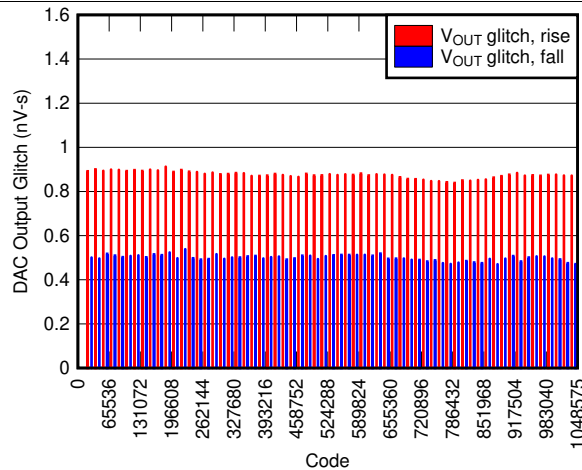
**Figure 31. Glitch Impulse, Rising Edge, 1-LSB Step**



**Figure 32. Glitch Impulse, Falling Edge, 1-LSB Step**

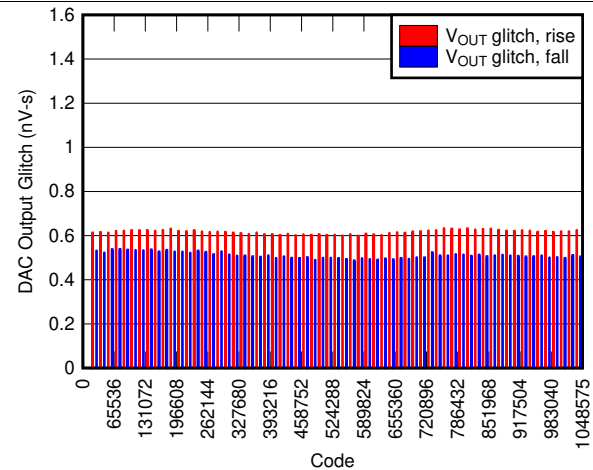
## Typical Characteristics (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)



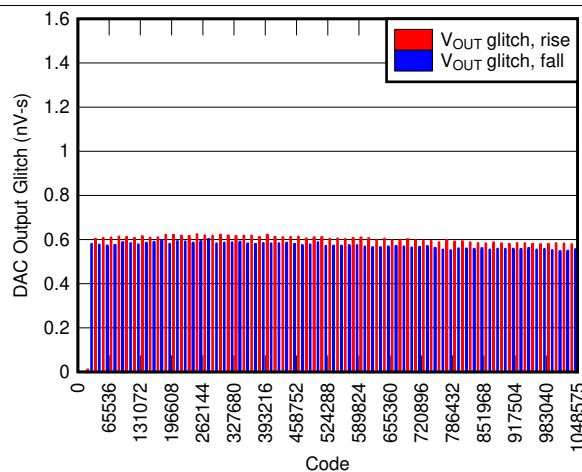
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = -10\text{ V}$

Figure 33. Segment Glitch Impulse, 1-LSB Step



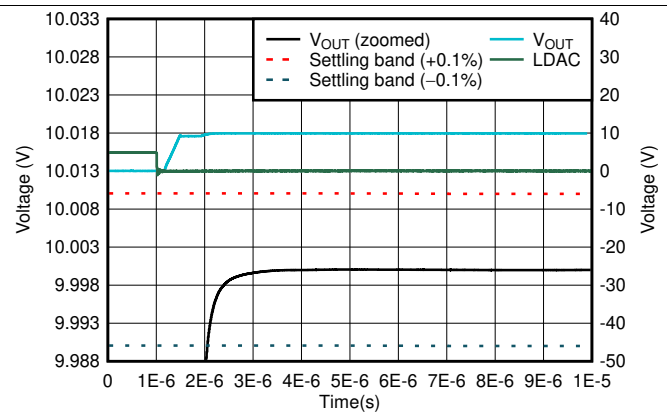
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 34. Segment Glitch Impulse, 1-LSB Step



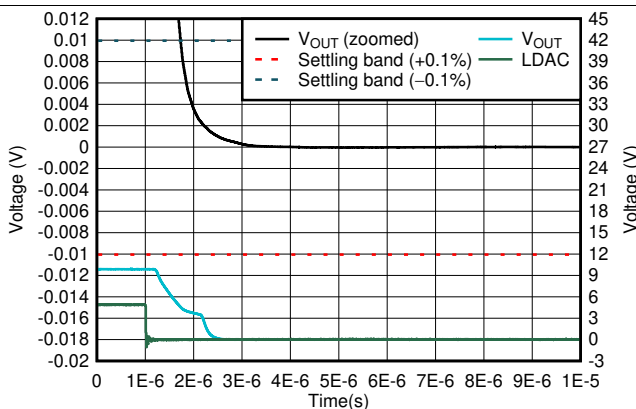
$V_{REFPF} = 5\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 35. Segment Glitch Impulse, 1-LSB Step



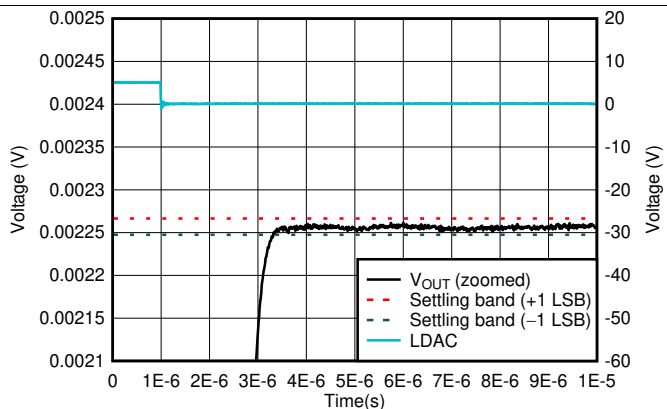
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 36. Full-Scale Settling Time, Rising Edge



$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$

Figure 37. Full-Scale Settling Time, Falling Edge

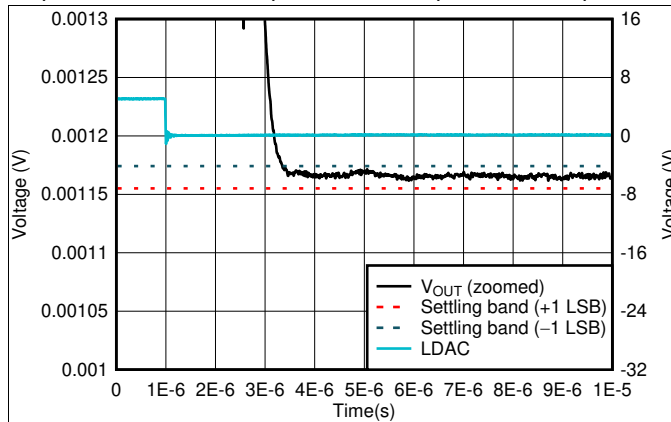


$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC transitions 100 codes around midscale

Figure 38. 100 Codes Settling Time, Rising Edge

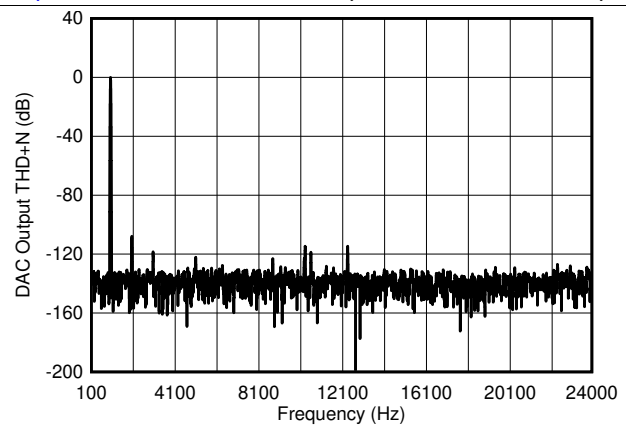
## Typical Characteristics (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_{CC} = 15\text{ V}$ ,  $V_{SS} = -15\text{ V}$ ,  $AV_{DD} = 5\text{ V}$ ,  $IOV_{DD} = 1.8\text{ V}$ , gain resistors unconnected (gain = 1x), OPA827 used as output and reference amplifier, UP = unipolar, BP = bipolar, and [temperature calibration](#) disabled (unless otherwise noted)



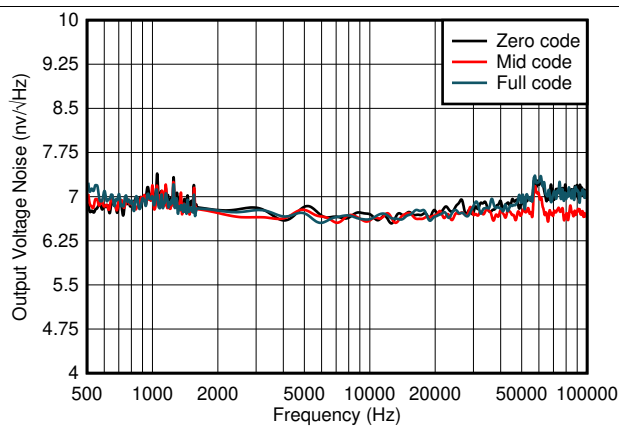
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC transitions 100 codes around midscale

**Figure 39. 100 Codes Settling Time, Falling Edge**



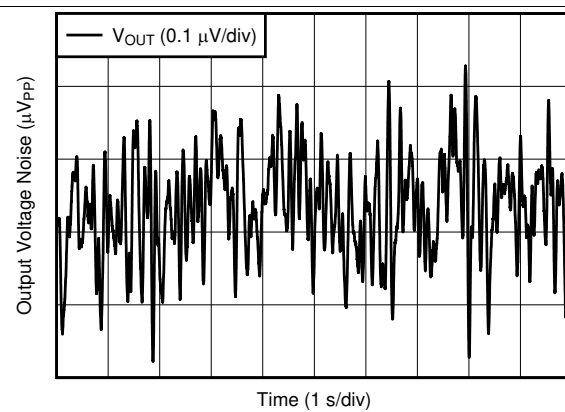
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC output frequency = 1 kHz, DAC update rate = 400 kHz

**Figure 40. Total Harmonic Distortion (THD + N) vs Frequency**



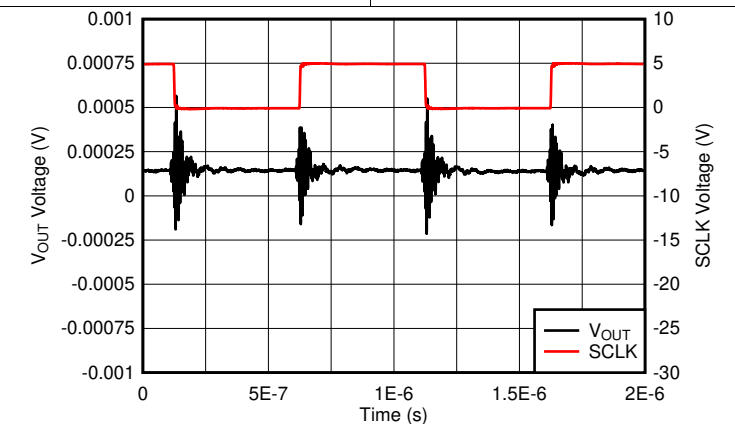
$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , measured at DAC output

**Figure 41. DAC Output Noise Spectral Density**



$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC at midcode, measured at DAC output pin

**Figure 42. DAC Output Noise: 0.1 Hz to 10 Hz**



$V_{REFPF} = 10\text{ V}$ ,  $V_{REFNF} = 0\text{ V}$ , DAC at midcode, measured at DAC output pin

**Figure 43. Clock Feedthrough**

## 8 Detailed Description

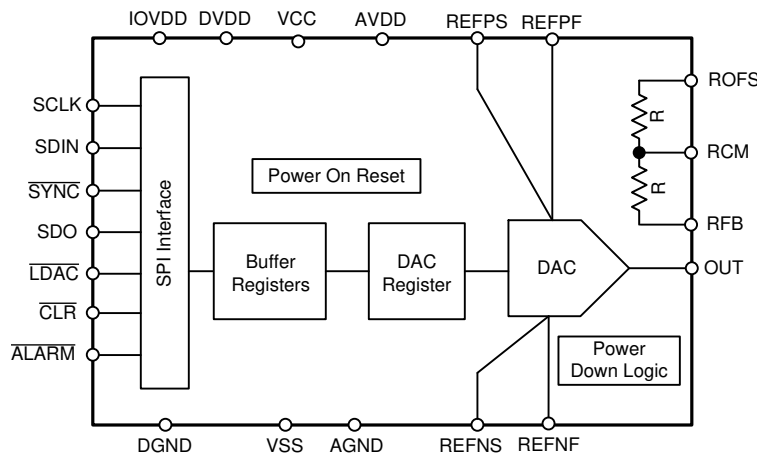
### 8.1 Overview

The 20-bit DAC11001A, 18-bit DAC91001, and 16-bit DAC81001 (DACx1001) are single-channel DACs. The unbuffered DAC output architecture is based on an R2R ladder that is designed to provide monotonicity over wide reference and temperature ranges (1-LSB DNL). This architecture provides a very low-noise (7 nV/√Hz) and fast-settling (1 μs) output. The DACx1001 also implement a deglitch circuit that enables low, code-independent glitch at the DAC output. This is extremely useful for creating ultra low harmonic distortion waveform generation.

The DACx1001 requires external reference voltages on REFPF and REFNF pins. The output of the DAC ranges from V<sub>REFNF</sub> to V<sub>REFPF</sub>. See the [Recommended Operating Conditions](#) for V<sub>REFPF</sub> and V<sub>REFNF</sub> voltage ranges.

The DACx1001 also includes precision matched gain setting pins (ROFS, RCM, and RFB). Using these pins and an external op amp, the DAC output can be scaled. The DACx1001 incorporate a power-on-reset circuit that makes sure that the DAC output powers up at zero scale, and remains at zero scale until a valid DAC command is issued. The DACx1001 use a 4-wire serial interface that operates at clock rates of up to 50 MHz.

### 8.2 Functional Block Diagram



### 8.3 Feature Description

#### 8.3.1 Digital-to-Analog Converter Architecture

The DACx1001 provide 20-bit monotonic outputs using an R2R ladder architecture. The DAC output ranges between V<sub>REFNF</sub> and V<sub>REFPF</sub> based on the 20-bit DAC data, as described in [Equation 1](#):

$$V_{OUT} = (V_{REFPF} - V_{REFNF}) \times \frac{CODE}{2^N} + V_{REFNF}$$

where

- CODE is the decimal equivalent of the DAC-DATA loaded to the DAC.
- N is the bits of resolution; 20 for DAC1101A, 18 for DAC91001, 16 for DAC81001.
- V<sub>REFPF</sub>, V<sub>REFNF</sub> is the reference voltage (positive and negative).

(1)

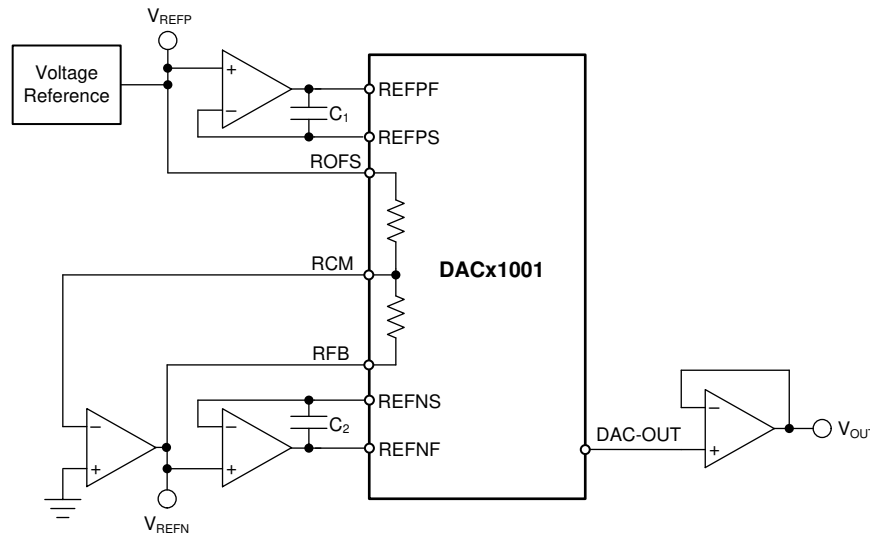
## Feature Description (continued)

### 8.3.2 External Reference

The DACx1001 require external references (REFPF and REFNF) to operate. See the [Recommended Operating Conditions](#) for  $V_{REFPF}$  and  $V_{REFNF}$  voltage ranges.

The DACx1001 also contain dedicated sense pins, REFPS for REFPF and REFNS for REFNF. The reference pins are unbuffered; therefore, use a reference driver circuit for these pins. Set the VREFVAL bits (address 02h) as per a reference span equal to  $(V_{REFPF} - V_{REFNF})$ . For example, the VREFVAL bits must be set to 0100 for  $V_{REFPF} = 5\text{ V}$  and  $V_{REFNF} = -5\text{ V}$ .

Figure 44 shows an example reference drive circuit for DACx1001. Table 1 shows the op-amp options for the reference driver circuit.



**Figure 44. Reference Drive Circuit**

**Table 1. Reference Op Amp Options**

| SELECTION PARAMETERS          | OP AMPS                |
|-------------------------------|------------------------|
| Low voltage and current noise | OPA211, OPA827, OPA828 |
| Low offset and drift          | OPA189                 |

### 8.3.3 Output Buffers

The DACx1001 outputs are unbuffered. Use an external op amp to buffer the DAC output. The DAC output voltage ranges from  $V_{REFPF}$  to  $V_{REFNF}$ . Two gain-setting resistors are integrated in the DACx1001. These resistors are used to scale the DAC output, minimize the bias current mismatch of the external op amp, and generate a negative reference for the REFNF pin. See the [Embedded Resistor Configurations](#) section for more information. Table 2 shows the op amp options for the output drive circuit.

**Table 2. Output Op Amp Options**

| SELECTION PARAMETERS      | OP AMPS                          |
|---------------------------|----------------------------------|
| Low bias current          | OPA827, OPA828                   |
| Low noise                 | OPA211, OPA828                   |
| Low offset and drift      | OPA189                           |
| Fast settling and low THD | OPA827, OPA828, OPA1612, THS4011 |

### 8.3.4 Internal Power-On Reset (POR)

The DACx1001 incorporate two internal POR circuits for the  $DV_{DD}$ ,  $AV_{DD}$ ,  $IOV_{DD}$ ,  $V_{CC}$ , and  $V_{SS}$  supplies. The POR signals are ANDed together, so that all supplies must be at the minimal specified values for the device to *not* be in a reset condition. These POR circuits initialize internal registers, as well as set the analog outputs to a known state while the device supplies are ramping. All registers are reset to default values. The DACx1001 power on with the DAC registers set to zero scale. The DAC can be powered down by writing 1 to PDN (bit 4, address 02h). Typically, the POR function can be ignored as long as the device supplies power up and maintain the specified minimum voltage levels. However, in the case of supply drop or brownout, the DACx1001 can have an internal POR reset event. Figure 45 represents the internal POR threshold levels for the  $DV_{DD}$ ,  $AV_{DD}$ ,  $IOV_{DD}$ ,  $V_{CC}$ , and  $V_{SS}$  supplies.

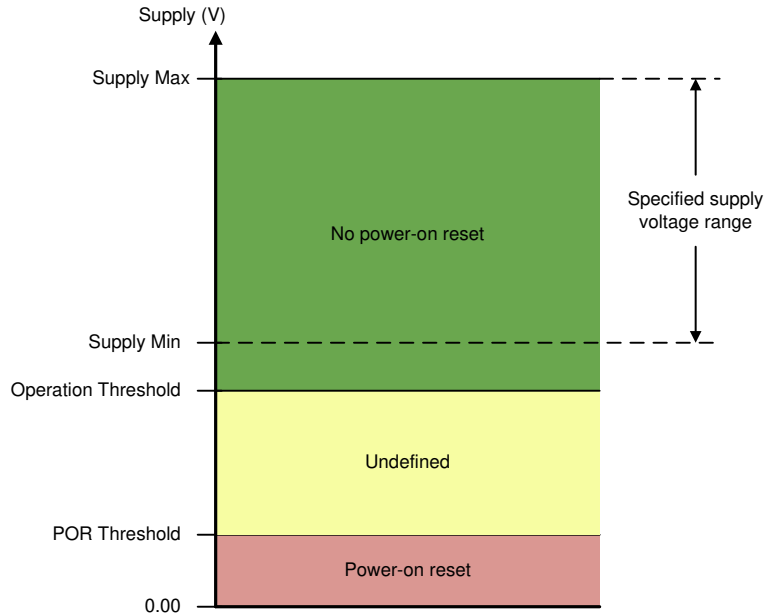


Figure 45. Relevant Voltage Levels for the POR Circuit

For the  $DV_{DD}$  supply, no internal POR occurs for nominal supply operation from 2.7 V (supply minimum) to 5.5 V (supply maximum). For a  $DV_{DD}$  supply region between 2.5 V (undefined operation threshold) and 1.6 V (POR threshold), the internal POR circuit may or may not provide a reset over all temperature conditions. For a  $DV_{DD}$  supply less than 1.6 V (POR threshold), the internal POR resets as long as the supply voltage is less than 1.6 V for approximately 1 ms.

For the  $AV_{DD}$  supply, no internal POR occurs for nominal supply operation from 4.5 V (supply minimum) to 5.5 V (supply maximum). For an  $AV_{DD}$  supply region between 4.1 V (undefined operation threshold) and 3.3 V (POR threshold), the internal POR circuit may or may not provide a reset over all temperature conditions. For an  $AV_{DD}$  supply less than 3.3 V (POR threshold), the internal POR resets as long as the supply voltage is less than 3.3 V for approximately 1 ms.

For the  $V_{CC}$  supply, no internal POR occurs for nominal supply operation from 8 V (supply minimum) to 36 V (supply maximum). For  $V_{CC}$  supply voltages between 7.5 V (undefined operation threshold) to 6 V (POR threshold), the internal POR circuit may or may not provide a reset over all temperature conditions. For a  $V_{CC}$  supply less than 6 V (POR threshold), the internal POR resets as long as the supply voltage is less than 6 V for approximately 1 ms.

For the  $V_{SS}$  supply, no internal POR occurs for nominal supply operation from –3 V (supply minimum) to –18 V (supply maximum). For  $V_{SS}$  supply voltages between –2.7 V (undefined operation threshold) to –1.8 V (POR threshold), the internal POR circuit may or may not provide a reset over all temperature conditions. For a  $V_{SS}$  supply greater than –1.8 V (POR threshold), the internal POR resets as long as the supply voltage is greater than –1.8 V for approximately 1 ms.

For the IOV<sub>DD</sub> supply, no internal POR occurs for nominal supply operation from 1.8 V (supply minimum) to 5.5 V (supply maximum). For IOV<sub>DD</sub> supply voltages between 1.5 V (undefined operation threshold) and 0.8 V (POR threshold), the internal POR circuit may or may not provide a reset over all temperature conditions. For an IOV<sub>DD</sub> supply less than 0.8 V (POR threshold), the internal POR resets as long as the supply voltage is less than 0.8 V for approximately 1 ms.

In case the DV<sub>DD</sub>, AV<sub>DD</sub>, IOV<sub>DD</sub>, V<sub>CC</sub>, or V<sub>SS</sub> supply drops to a level where the internal POR signal is indeterminate, power cycle the device followed by a software reset.

### 8.3.5 Temperature Drift and Calibration

The DACx1001 includes a calibration circuit that significantly reduces the temperature drift on integrated and differential nonlinearities. By default, this feature is disabled. Enable the temperature calibration feature by writing 1 to the EN\_TMP\_CAL bit (address 02h, B23). After the EN\_TMP\_CAL bit is set, issue a calibration cycle by writing 1 to RCLTMP (address 04h, B8). At this point, the device enters a calibration cycle. Do not issue any DAC update command during this period. The device has the capability to indicate the end of calibration using two methods:

1. Read the status bit ALM (address 05h, B12) using SPI.
2. Issue an alarm on the ALARM pin by setting logic 0. To enable this feature, write 1 to ENALMP bit (address 02h, B12).

After the calibration cycle completes, update the DAC code to observe the impact at the DAC output. If the environmental temperature changes after calibration, then recalibrate the device.

### 8.3.6 DAC Output Deglitch Circuit

The DACx1001 include a deglitch (track-and-hold) circuit at the output. This circuit is enabled by default. The deglitch circuit minimizes the code-to-code glitch at the DAC output at the expense of the DAC update rate. This circuit is disabled by writing 1 to DIS\_TNH (bit 7, address 06h). Disable this circuit to enable faster update of the DAC output, but with higher code-to-code glitches.

## 8.4 Device Functional Modes

### 8.4.1 Fast-Settling Mode and THD

The DACx1001 R2R ladder and deglitch circuit reduce the harmonic distortion for waveform generation applications. The fast settling bit (FSET, bit 10, address 02h) is set to 1 by default, so that the DAC is configured for enhanced THD performance. The FSET bit can be reset to 0 using an SPI write to enable fast-settling mode. In this mode, the DAC deglitcher circuit can be configured using TNH\_MASK (bits 19:18, address 02h). These bits disable the deglitch circuit for code changes specified in [Table 7](#). These bits are only writable when FSET = 0 (fast settling enabled) and DIS\_TNH = 0 (deglitch circuit enabled).

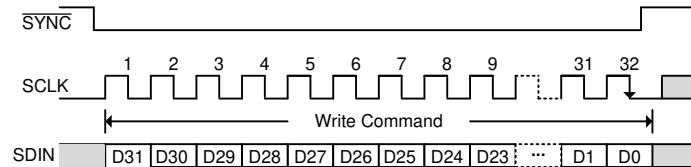
### 8.4.2 DAC Update Rate Mode

The DACx1001 maximum update rate can be configured up to 1 MHz by using UP\_RATE (bits 6:4, address 06h). These bits change the hold timing of the deglitch circuit. The bits are set to a 0.5-MHz DAC update rate by default for enhanced THD performance. Changing the maximum update rate of the DAC impacts THD performance.

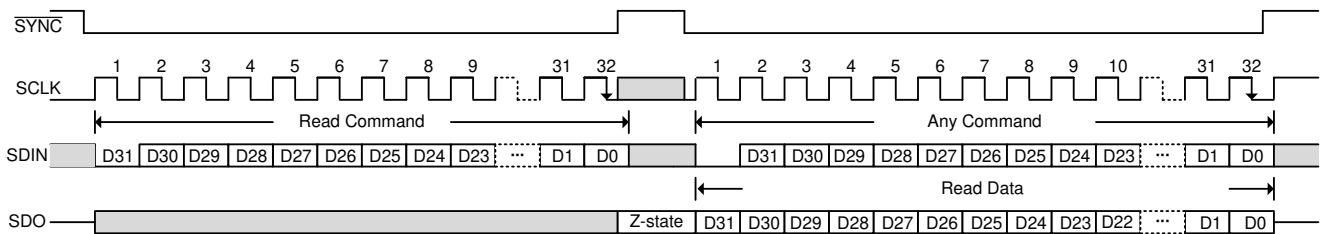
## 8.5 Programming

The DACx1001 family of devices is controlled through a flexible four-wire serial interface that is compatible with serial interfaces used on many microcontrollers and DSP controllers. The interface provides read and write access to all registers of the DACx1001 devices. Additionally, the interface can be configured to daisy-chain multiple devices for write operations.

Each serial interface access cycle is exactly 32 bits long, as shown in Figure 46. A frame is initiated by asserting  $\overline{\text{SYNC}}$  pin low. The frame ends when the  $\overline{\text{SYNC}}$  pin is deasserted high. The first bit is read/write bit B31. A write is performed when this bit is set to 0, and a read is performed when this bit is set to 1. The next 7 bits are address bits B30 to B24. The next 20 bits are data. For all writes, data are clocked on the falling edge of SCLK. As Figure 47 shows, for read access and daisy-chain operation, the data are clocked out on the SDO terminal on the rising edge of SCLK.



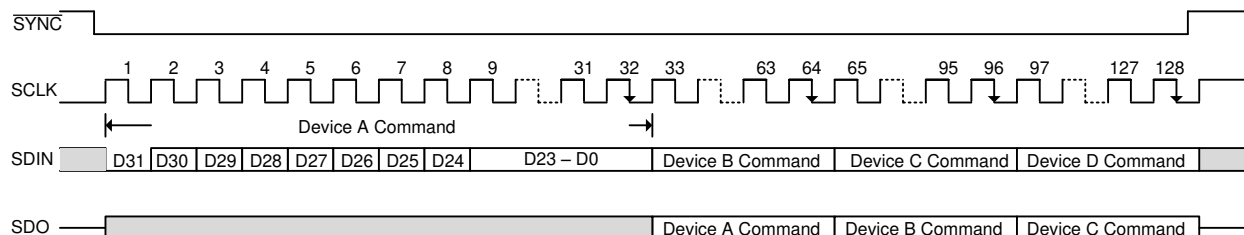
**Figure 46. Serial Interface Write Bus Cycle: Standalone Mode**



**Figure 47. Serial Interface Read Bus Cycle**

### 8.5.1 Daisy-Chain Operation

For systems that contain several DACx1001 devices, the SDO pin is used to daisy-chain the devices together. The daisy-chain feature is useful in reducing the number of serial interface lines. The first falling edge on the  $\overline{\text{SYNC}}$  pin starts the operation cycle, as shown in Figure 48. SCLK is continuously applied to the input shift register while the  $\overline{\text{SYNC}}$  pin is kept low. The DAC is updated with the data on rising edge of  $\overline{\text{SYNC}}$  pin.



**Figure 48. Serial Interface Daisy-Chain Write Cycle**

If more than 32 clock pulses are applied, the data ripple out of the shift register and appear on the SDO line. These data are clocked out on the rising edge of SCLK and are valid on the falling edge. By connecting the SDO output of the first device to the SDI input of the next device in the chain, a multiple-device interface is constructed. Each device in the system requires 32 clock pulses.

## Programming (continued)

As a result, the total number of clock cycles must be equal to  $32 \times N$ , where  $N$  is the total number of devices in the daisy-chain. When the serial transfer to all devices is complete the  $\overline{\text{SYNC}}$  signal is taken high. This action transfers the data from the SPI shift registers to the internal register of each device in the daisy-chain and prevents any further data from being clocked into the input shift register. The DACx1001 implement a bit that enables higher speeds for clocking out data from the SDO pin. Enable this feature by setting FSDO (bit 13, address 02h) to 1. See [Timing Requirements: Read and Daisy-Chain Write,  \$2.7\text{ V} \leq \text{DV}\_{\text{DD}} < 4.5\text{ V}\$](#)  and [Timing Requirements: Read and Daisy-Chain Write,  \$4.5\text{ V} \leq \text{DV}\_{\text{DD}} \leq 5.5\text{ V}\$](#)  for more information.

### 8.5.2 $\overline{\text{CLR}}$ Pin Functionality and Software Clear

The  $\overline{\text{CLR}}$  pin is an asynchronous input pin to the DAC. When activated, this level-sensitive pin clears the DAC buffers and DAC latches to the DAC-CLEAR-DATA bits (address 03h). The device exits clear mode on the  $\overline{\text{SYNC}}$  rising edge of the next valid write to the device. If the  $\overline{\text{CLR}}$  pin receives a logic 0 during a write sequence during normal operation, the clear mode is activated and the buffer and DAC registers are immediately cleared. The DAC registers can also be cleared using the SCLR bit (address 04h, B5); the contents are cleared at the rising edge of  $\overline{\text{SYNC}}$ .

### 8.5.3 Output Update (Synchronous and Asynchronous)

The DACx1004 devices offer both a software and hardware simultaneous update and control function. The DAC double-buffered architecture has been designed so that new data can be entered for the DAC without disturbing the analog output. Data updates can be performed either in synchronous or in asynchronous mode, depending on the status of LDAC-MODE bit (address 02h, B14).

#### 8.5.3.1 Synchronous Update

In synchronous mode (LDACMODE = 1), the  $\overline{\text{LDAC}}$  pin is used as an active-low signal for simultaneous DAC updates. Data buffers must be loaded with the desired data before an  $\overline{\text{LDAC}}$  low pulse. After an  $\overline{\text{LDAC}}$  low pulse, the DAC is updated with the last contents of the corresponding data buffers. If the content of a data buffer is not changed, the DAC output remains unchanged after the  $\overline{\text{LDAC}}$  pin is pulsed low.

#### 8.5.3.2 Asynchronous Update

In asynchronous mode (LDACMODE = 0), data are updated with the rising edge of the  $\overline{\text{SYNC}}$  (when daisy-chain mode is enabled, DSDO = 0), or at the 32nd falling edge of SCLK (When daisy-chain mode is disabled, DSDO = 1). For asynchronous updates, the  $\overline{\text{LDAC}}$  pin is not required, and must be connected to 0 V permanently.

### 8.5.4 Software Reset Mode

The DACx1001 implements a software reset feature. The software reset function uses the SRST bit (address 04h, B6). When this bit is set to 1, the device resets to the default state.

## 8.6 Register Map

**Table 3. Register Map**

| REGISTER NAME  | BIT              |       |                                                         |    |    |    |          |    |    |    |    |           |      |        |      |      |         |         |   |   |      |             |     |    |    |    |
|----------------|------------------|-------|---------------------------------------------------------|----|----|----|----------|----|----|----|----|-----------|------|--------|------|------|---------|---------|---|---|------|-------------|-----|----|----|----|
|                | 31               | 30-24 | 23                                                      | 22 | 21 | 20 | 19       | 18 | 17 | 16 | 15 | 14        | 13   | 12     | 11   | 10   | 9       | 8       | 7 | 6 | 5    | 4           | 3-0 |    |    |    |
| NOP            | $\overline{W}$   | 00h   | NOP                                                     |    |    |    |          |    |    |    |    |           |      |        |      |      |         |         |   |   |      |             |     | 0h |    |    |
| DAC-DATA       | $R/\overline{W}$ | 01h   | DAC-DATA (20 bits, 18 bits, or 16 bits, left-justified) |    |    |    |          |    |    |    |    |           |      |        |      |      |         |         |   |   |      |             |     | 0h |    |    |
| CONFIG1        | $R/\overline{W}$ | 02h   | EN_TMP_CAL                                              | 0h |    |    | TNH_MASK |    | 0h |    |    | LDAC MODE | FSDO | ENALMP | DSDO | FSET | VREFVAL |         |   |   | 0    | PDN         | 0h  |    |    |    |
| DAC-CLEAR-DATA | $R/\overline{W}$ | 03h   | DAC-CLEAR-DATA (8 bits left justified)                  |    |    |    |          |    |    |    |    | 000h      |      |        |      |      |         |         |   |   |      |             |     |    | 0h |    |
| TRIGGER        | $R/\overline{W}$ | 04h   | 0000h                                                   |    |    |    |          |    |    |    |    |           |      |        |      |      |         | RCLTMP  |   | 0 | SRST | SCLR        | 0   | 0h |    |    |
| STATUS         | R                | 05h   | 000h                                                    |    |    |    |          |    |    |    |    |           |      | ALM    |      | 00h  |         |         |   |   |      |             |     |    |    | 0h |
| CONFIG2        | $R/\overline{W}$ | 06h   | 0000h                                                   |    |    |    |          |    |    |    |    |           |      |        |      |      |         | DIS_TNH |   |   |      | TNH_SETTING |     |    |    | 0h |

**Table 4. Access Type Codes**

| Access Type                   | Code | Description                            |
|-------------------------------|------|----------------------------------------|
| <b>Read Type</b>              |      |                                        |
| R                             | R    | Read                                   |
| <b>Write Type</b>             |      |                                        |
| W                             | W    | Write                                  |
| <b>Reset or Default Value</b> |      |                                        |
| -n                            |      | Value after reset or the default value |

### 8.6.1 NOP Register (address = 00h) [reset = 0x000000h]

**Figure 49. NOP Register Format**

|                |    |         |    |    |    |    |    |    |    |     |    |    |    |    |    |
|----------------|----|---------|----|----|----|----|----|----|----|-----|----|----|----|----|----|
| 31             | 30 | 29      | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21  | 20 | 19 | 18 | 17 | 16 |
| Read/Write     |    | Address |    |    |    |    |    |    |    | NOP |    |    |    |    |    |
| $\overline{W}$ |    | W       |    |    |    |    |    |    |    | W   |    |    |    |    |    |
| 15             | 14 | 13      | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5   | 4  | 3  | 2  | 1  | 0  |
| NOP            |    |         |    |    |    |    |    |    |    |     |    | 0h |    |    |    |
| W              |    |         |    |    |    |    |    |    |    |     |    | W  |    |    |    |

**Table 5. NOP Register Field Descriptions**

| Bit   | Field   | Type | Reset  | Description                 |
|-------|---------|------|--------|-----------------------------|
| 31    | Write   | W    | N/A    | Write when set to 0         |
| 30:24 | Address | W    | N/A    | 00h                         |
| 23:4  | NOP     | W    | 00000h | No operation - Write 00000h |
| 3:0   | 0h      | W    | N/A    | N/A                         |

### 8.6.2 DAC-DATA Register (address = 01h) [reset = 0x000000h]

**Figure 50. DAC-DATA Register Format**

|                                                      |    |         |    |    |    |    |    |    |    |                                                      |    |    |    |    |    |
|------------------------------------------------------|----|---------|----|----|----|----|----|----|----|------------------------------------------------------|----|----|----|----|----|
| 31                                                   | 30 | 29      | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21                                                   | 20 | 19 | 18 | 17 | 16 |
| Read/Write                                           |    | Address |    |    |    |    |    |    |    | DAC-DATA (20-bit, 18-bit, or 16-bit, left justified) |    |    |    |    |    |
| R/ $\overline{W}$                                    |    | W       |    |    |    |    |    |    |    | R/W                                                  |    |    |    |    |    |
| 15                                                   | 14 | 13      | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5                                                    | 4  | 3  | 2  | 1  | 0  |
| DAC-DATA (20-bit, 18-bit, or 16-bit, left justified) |    |         |    |    |    |    |    |    |    |                                                      |    | 0h |    |    |    |
| R/W                                                  |    |         |    |    |    |    |    |    |    |                                                      |    | W  |    |    |    |

**Table 6. DAC-DATA Register Field Descriptions**

| Bit   | Field          | Type              | Reset | Description                                                                                                                                                                                                                                              |
|-------|----------------|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write     | R/ $\overline{W}$ | N/A   | Read when set to 1 or write when set to 0                                                                                                                                                                                                                |
| 30:24 | Address        | W                 | N/A   | 01h                                                                                                                                                                                                                                                      |
| 23:4  | DAC-DATA[19:0] | R/W               | 0h    | Stores the 20-bit, 18-bit, or 16-bit data to be loaded to DAC in MSB aligned straight binary format.<br>Data follows the format below:<br>DAC1101A: { DAC-DATA[19:0] }<br>DAC91001: { DAC-DATA[17:0], 0, 0 }<br>DAC81001: { DAC-DATA[15:0], 0, 0, 0, 0 } |
| 3:0   | 0h             | W                 | N/A   | N/A                                                                                                                                                                                                                                                      |

### 8.6.3 CONFIG1 Register (address = 02h) [reset = 004C80h for bits [23:0]]

**Figure 51. CONFIG1 Register Format**

| 31             | 30           | 29   | 28     | 27   | 26   | 25      | 24 | 23                 | 22  | 21 | 20       | 19  | 18  | 17 | 16 |  |
|----------------|--------------|------|--------|------|------|---------|----|--------------------|-----|----|----------|-----|-----|----|----|--|
| Read/<br>Write | Address      |      |        |      |      |         |    | EN_<br>TMP_<br>CAL | 0h  |    | TNH_MASK |     | 0h  |    |    |  |
| R/W            |              | W    |        |      |      |         |    |                    | R/W |    | W        |     | R/W |    | W  |  |
| 15             | 14           | 13   | 12     | 11   | 10   | 9       | 8  | 7                  | 6   | 5  | 4        | 3   | 2   | 1  | 0  |  |
| 0h             | LDAC<br>MODE | FSDO | ENALMP | DSDO | FSET | VREFVAL |    |                    |     | 0h | PDN      | 0h  |     |    |    |  |
| W              |              |      |        |      | R/W  |         |    |                    |     | W  |          | R/W |     | W  |    |  |

**Table 7. CONFIG1 Register Field Descriptions**

| Bit   | Field      | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write | R/W  | N/A   | Read when set to 1 or write when set to 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 30:24 | Address    | W    | N/A   | 02h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 23    | EN_TMP_CAL | R/W  | 0h    | Enables and disables the temperature calibration feature<br>0 : Temperature calibration feature disabled (default)<br>1 : Temperature calibration feature enabled                                                                                                                                                                                                                                                                                                                                                                     |
| 22:20 | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 19-18 | TNH_MASK   | R/W  | 0h    | Mask track and hold (TNH) circuit. This bit is writable only when FSET = 0 [fast-settling mode] and DIS_TNH = 0 [track-and-hold enabled]<br>00: TNH masked for code jump > 2 <sup>14</sup> (default)<br>01: TNH masked for code jump > 2 <sup>15</sup><br>10: TNH masked for code jump > 2 <sup>13</sup><br>11: TNH masked for code jump > 2 <sup>12</sup>                                                                                                                                                                            |
| 17:15 | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 14    | LDACMODE   | R/W  | 1     | Synchronous or asynchronous mode select bit<br>0 : DAC output updated on SYNC rising edge<br>1 : DAC updated on LDAC falling edge (default)                                                                                                                                                                                                                                                                                                                                                                                           |
| 13    | FSDO       | R/W  | 0h    | Enable Fast SDO<br>0 : Fast SDO disabled (Default)<br>1 : Fast SDO enabled                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 12    | ENALMP     | R/W  | 0h    | Enable ALARM pin to be pulled low, end of temperature calibration cycle<br>0 : No alarm on the ALARM pin<br>1 : Indicates end of temperature calibration cycle. ALARM pin pulled low.                                                                                                                                                                                                                                                                                                                                                 |
| 11    | DSDO       | R/W  | 1h    | Enable SDO (for readback and daisy-chain)<br>1 : SDO enabled (default)<br>0 : SDO disabled                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10    | FSET       | R/W  | 1h    | Fast-settling vs enhanced THD mode<br>0 : Fast settling<br>1 : Enhanced THD (default)                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 9:6   | VREFVAL    | R/W  | 2h    | Reference span value bits<br>0000: Invalid<br>0001: Invalid<br>0010: Reference span = 5 V ± 1.25 V (default)<br>0011: Reference span = 7.5 V ± 1.25 V<br>0100: Reference span = 10 V ± 1.25 V<br>0101: Reference span = 12.5 V ± 1.25 V<br>0110: Reference span = 15 V ± 1.25 V<br>0111: Reference span = 17.5 V ± 1.25 V<br>1000: Reference span = 20 V ± 1.25 V<br>1001: Reference span = 22.5 V ± 1.25 V<br>1010: Reference span = 25 V ± 1.25 V<br>1011: Reference span = 27.5 V ± 1.25 V<br>1100: Reference span = 30 V ± 1.25 V |
| 5     | 0          | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4     | PDN        | R/W  | 0h    | Powers down and power up the DAC<br>0 : DAC power up (default)<br>1 : DAC power down                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3:0   | 0000       | R/W  | N/A   | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 8.6.4 DAC-CLEAR-DATA Register (address = 03h) [reset = 000000h for bits [23:0]]

**Figure 52. DAC-CLEAR-DATA Register Format**

| 31         | 30      | 29 | 28 | 27 | 26 | 25 | 24 | 23                                      | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|------------|---------|----|----|----|----|----|----|-----------------------------------------|----|----|----|----|----|----|----|
| Read/Write | Address |    |    |    |    |    |    | DAC-CLEAR-DATA (8 bits, left justified) |    |    |    |    |    |    |    |
| R/W        | W       |    |    |    |    |    |    | R/W                                     |    |    |    |    |    |    |    |
| 15         | 14      | 13 | 12 | 11 | 10 | 9  | 8  | 7                                       | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| 000h       |         |    |    |    |    |    |    |                                         |    |    |    | 0h |    |    |    |
| W          |         |    |    |    |    |    |    |                                         |    |    |    | W  |    |    |    |

**Table 8. DAC-CLEAR-DATA Register Field Descriptions**

| Bit   | Field          | Type | Reset | Description                                                                                                                                               |
|-------|----------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write     | R/W  | N/A   | Read when set to 1 or write when set to 0                                                                                                                 |
| 30:24 | Address        | W    | N/A   | 03h                                                                                                                                                       |
| 23:16 | DAC-CLEAR-DATA | R/W  | 00h   | Stores the 8-bit data to be loaded to DAC in left-justified, straight-binary format. DAC data registers updated with this value when CLR pin asserted low |
| 15:0  | 000h           | W    | N/A   | N/A                                                                                                                                                       |

## 8.6.5 TRIGGER Register (address = 04h) [reset = 000000h for bits [23:0]]

**Figure 53. TRIGGER Register Format**

| 31         | 30      | 29 | 28 | 27 | 26 | 25 | 24     | 23  | 22   | 21   | 20 | 19 | 18 | 17 | 16 |
|------------|---------|----|----|----|----|----|--------|-----|------|------|----|----|----|----|----|
| Read/Write | Address |    |    |    |    |    |        | 00h |      |      |    |    |    |    |    |
| R/W        | W       |    |    |    |    |    |        | W   |      |      |    |    |    |    |    |
| 15         | 14      | 13 | 12 | 11 | 10 | 9  | 8      | 7   | 6    | 5    | 4  | 3  | 2  | 1  | 0  |
| 00h        |         |    |    |    |    |    | RCLTMP | 0h  | SRST | SCLR | 0h | 0h |    |    |    |
| W          |         |    |    |    |    |    | R/W    | W   | R/W  | R/W  | W  | W  |    |    |    |

**Table 9. TRIGGER Register Field Descriptions**

| Bit   | Field      | Type | Reset | Description                                                                                                                                                                                                                                                                                   |
|-------|------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write | R/W  | N/A   | Read when set to 1 or write when set to 0                                                                                                                                                                                                                                                     |
| 30:24 | Address    | W    | N/A   | 04h                                                                                                                                                                                                                                                                                           |
| 23:9  | 0000h      | W    | N/A   | Unused                                                                                                                                                                                                                                                                                        |
| 8     | RCLTMP     | R/W  | 0h    | Trigger temperature recalibration DAC Codes<br>0 : No temperature recalibration (default)<br>1 : DAC codes recalibrated, ALARM pin is pulled low (if ENALMP = 1) and ALM bit (Address 05) is set 1 upon calibration completion. Subsequent DAC codes will use latest calibrated coefficients. |
| 7     | 0h         | W    | N/A   | NA                                                                                                                                                                                                                                                                                            |
| 6     | SRST       | R/W  | 0h    | Software reset<br>0 : No software reset (default)<br>1 : Software reset initiated, device in default state                                                                                                                                                                                    |
| 5     | SCLR       | R/W  | 0h    | Software clear<br>0 : No software clear (default)<br>1 : Software clear initiated, DAC registers in clear mode, DAC code set by clear select register (address 03h). DAC output clears on 32nd SCLK falling (DSDO = 1) or SYNC rising edge (DSDO = 0)                                         |
| 4     | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                           |
| 3:0   | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                           |

## 8.6.6 STATUS Register (address = 05h) [reset = 000000h for bits [23:0]]

**Figure 54. STATUS Register Format**

| 31             | 30      | 29 | 28  | 27  | 26 | 25 | 24 | 23  | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|----------------|---------|----|-----|-----|----|----|----|-----|----|----|----|----|----|----|----|
| Read/<br>Write | Address |    |     |     |    |    |    | 00h |    |    |    |    |    |    |    |
| R              | W       |    |     |     |    |    |    | W   |    |    |    |    |    |    |    |
| 15             | 14      | 13 | 12  | 11  | 10 | 9  | 8  | 7   | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| 0h             |         |    | ALM | 00h |    |    |    |     |    |    |    | 0h |    |    |    |
| W              |         |    | R   | W   |    |    |    |     |    |    |    | W  |    |    |    |

**Table 10. STATUS Register Field Descriptions**

| Bit   | Field      | Type | Reset | Description                                                                                                                                                                                                                                                                                         |
|-------|------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write | R    | N/A   | Read when set to 1 , read only                                                                                                                                                                                                                                                                      |
| 30:24 | Address    | W    | N/A   | 05h                                                                                                                                                                                                                                                                                                 |
| 23:13 | 000h       | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                 |
| 12    | ALM        | R    | 0     | Alarm indicator bit, This bit is not masked by ENALMP bit<br>0 :Temperature recalibration in progress<br>1 : DAC codes recalibrated, ALARM pin is pulled low (if ENALMP = 1) Subsequent DAC codes will use latest calibrated coefficients. Reading back this register resets ALARM pin to 1 status. |
| 11:4  | 00h        | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                 |
| 3:0   | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                 |

## 8.6.7 CONFIG2 Register (address = 06h) [reset = 000040h for bits [23:0]]

**Figure 55. CONFIG2 Register Format**

| 31             | 30      | 29 | 28 | 27 | 26 | 25 | 24 | 23      | 22      | 21 | 20 | 19 | 18 | 17 | 16 |
|----------------|---------|----|----|----|----|----|----|---------|---------|----|----|----|----|----|----|
| Read/<br>Write | Address |    |    |    |    |    |    | 00h     |         |    |    |    |    |    |    |
| R/W            | W       |    |    |    |    |    |    | W       |         |    |    |    |    |    |    |
| 15             | 14      | 13 | 12 | 11 | 10 | 9  | 8  | 7       | 6       | 5  | 4  | 3  | 2  | 1  | 0  |
| 00h            |         |    |    |    |    |    |    | DIS_TNH | UP_RATE |    |    | 0h |    |    |    |
| W              |         |    |    |    |    |    |    | R/W     | R/W     |    |    | W  |    |    |    |

**Table 11. CONFIG2 Register Field Descriptions**

| Bit   | Field      | Type | Reset | Description                                                                                                                                                                                                                                                                                                   |
|-------|------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Read/Write | R/W  | N/A   | Read when set to 1 or write when set to 0                                                                                                                                                                                                                                                                     |
| 30:24 | Address    | W    | N/A   | 06h                                                                                                                                                                                                                                                                                                           |
| 23:8  | 0000h      | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                           |
| 7     | DIS_TNH    | R/W  | 0h    | Disable track and hold:<br>0 : Track and hold enabled (default)<br>1 : Track and hold disabled                                                                                                                                                                                                                |
| 6-4   | UP_RATE    | R/W  | 4h    | DAC output max update rate:<br>000: 1 MHz with 38-MHz SCLK<br>001: 0.9 MHz with 34-MHz SCLK<br>010: 0.8 MHz with 31-MHz SCLK<br>011: 1.2 MHz with 45-MHz SCLK<br>100: 0.5 MHz with 21-MHz SCLK, (default)<br>101: 0.45 MHz with 18-MHz SCLK<br>110: 0.4 MHz with 16-MHz SCLK<br>111: 0.6 MHz with 24-MHz SCLK |
| 3:0   | 0h         | W    | N/A   | N/A                                                                                                                                                                                                                                                                                                           |

## 9 Application and Implementation

### NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

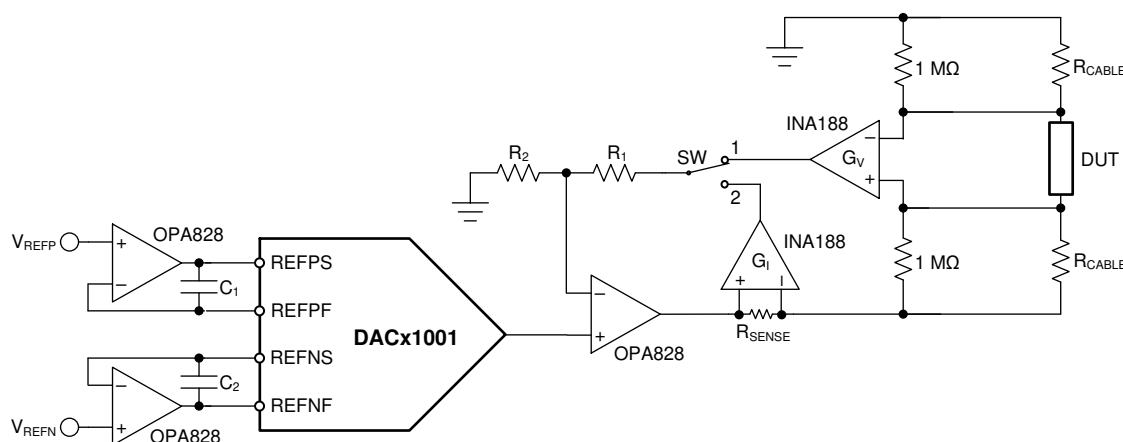
### 9.1 Application Information

The DACx1001 family of DACs are targeted for high-precision applications where ultra-high dc accuracy, ultra-low noise, fast settling, or high total harmonic distortion (THD) is required. The DACx1001 provides 20-bit monotonic resolution. This device finds application in high-performance source measure unit (SMU), battery test equipment (BTE), arbitrary waveform generation (AWG), and closed-loop control applications such as microelectromechanical system (MEMS) actuators, linear actuators, precision motor control, lens autofocus control in precision microscopy, lens control in mass spectrometer, beam control in electron beam lithography, and so on.

### 9.2 Typical Application

#### 9.2.1 Source Measure Unit (SMU)

A source measure unit (SMU) is a common building block in memory and semiconductor test equipment and bench-top source measure units. A DAC is used in an SMU to force a desired voltage or a current to a device-under-test (DUT). [Figure 56](#) provides a simplified circuit diagram of the force-DAC in an SMU.



**Figure 56. Source Measure Unit**

#### 9.2.1.1 Design Requirements

- Force voltage range:  $\pm 10$  V
- Force current range:  $\pm 20$  mA

#### 9.2.1.2 Detailed Design Procedure

The DAC11001A is an excellent choice for this application to meet the 20-bit resolution requirement. Switch SW is used to toggle between force-voltage and force-current modes, as shown in [Figure 56](#). The [OPA828](#) is a high-precision amplifier that provides a good balance between dc and ac performance, and can supply  $\pm 30$ -mA output current. The [INA188](#) is a zero-drift instrumentation amplifier with gain selected with an external resistor. The external resistor is not shown in the drawing for simplicity. The gain resistor is not required for a gain of 1. [Equation 2](#) shows the calculation of the voltage gain when switch SW is in position 1.

$$A_V = \frac{1}{G_V} \times \left( 1 + \frac{R_1}{R_2} \right)$$

(2)

## Typical Application (continued)

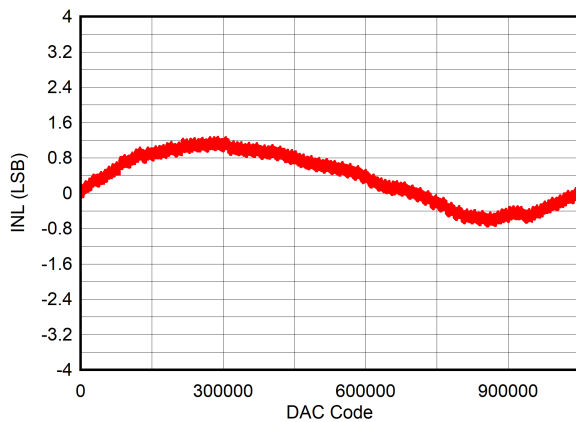
Precision reference sources are available at 5 V or less. Use a  $\pm 5$ -V reference with a 2x gain configuration to get an output of  $\pm 10$  V. The DAC output amplifier sets the gain at 2, assuming  $G_V = 1$ , as shown in Equation 3.  $R_1$  and  $R_2$  are 1-k $\Omega$  each. Equation 3 shows the calculation for the current gain when the switch is in the position 2.

$$A_V = \frac{1}{R_{SENSE} \times G_I} \times \left( 1 + \frac{R_1}{R_2} \right) \quad (3)$$

In order to get  $\pm 20$ -mA output current range with  $R_1 = R_2$ ,  $R_{SENSE} \times G_I$  must be 500. Choose  $G_I$  as 50 so that  $R_{SENSE}$  can be 10- $\Omega$ . For a  $\pm 20$ -mA output current, the voltage drop across  $R_{SENSE}$  is  $\pm 200$ -mV. In case the design requires a lower voltage headroom, choose a higher value for  $G_I$  and a smaller resistance value for  $R_{SENSE}$ .

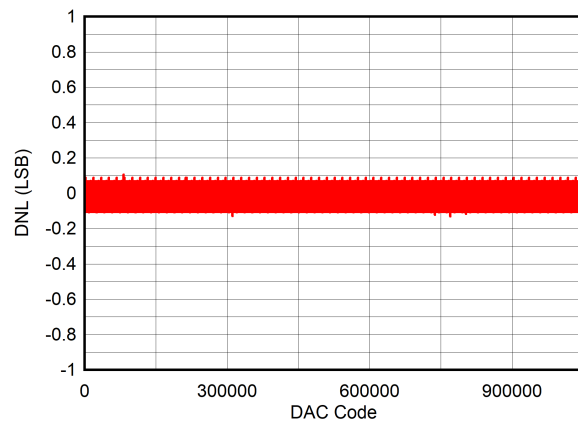
There is no equation to select  $C_1$  and  $C_2$ . The values of  $C_1$  and  $C_2$  depend on the stability criteria of the reference buffers when driving the reference inputs of DACx1001. The values are obtained through simulation. For the OPA828, use  $C_1 = C_2 = 100$  pF. The 1-M $\Omega$  resistors in the circuit are used for making sure the amplifiers are not left in an open-loop state.

### 9.2.1.3 Application Curves



Measured on BP-DAC11001EVM, external 10-V reference source

**Figure 57. INL at  $\pm 10$ -V Output**



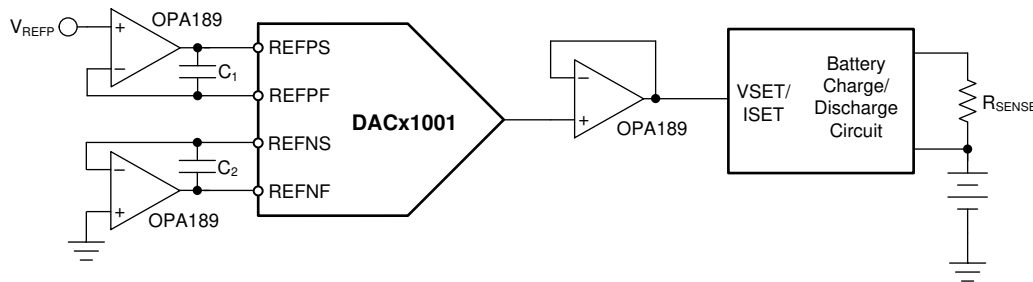
Measured on BP-DAC11001EVM, external 10-V reference source

**Figure 58. DNL at  $\pm 10$ -V Output**

## Typical Application (continued)

### 9.2.2 Battery Test Equipment (BTE)

Battery test equipment is used for lithium-ion battery formation, end-of-line testing, and diagnostics. For battery diagnostics, high-precision DACs, such as the DACx1001, are required to maintain a highly stable voltage over temperature and time.



**Figure 59. Battery Test Equipment**

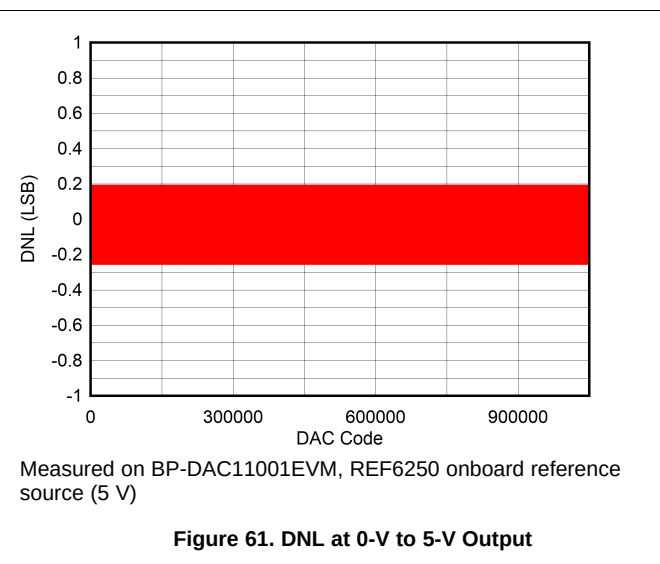
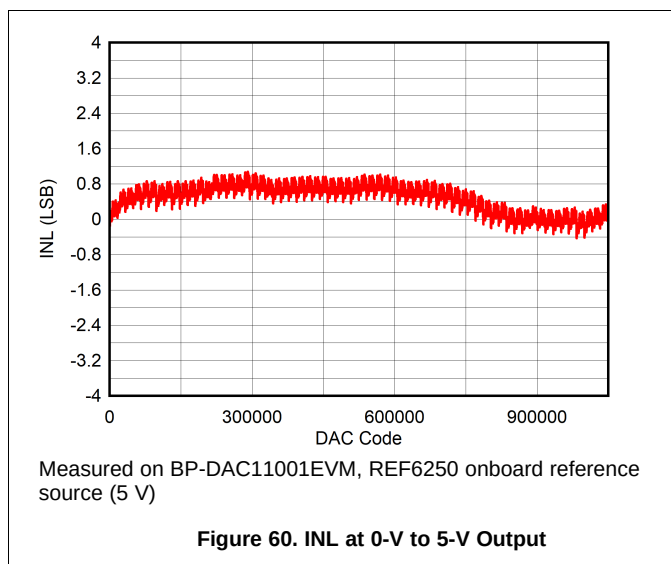
#### 9.2.2.1 Design Requirements

- Output range: 0 V to 5 V
- System level temperature drift:  $\pm 2$  ppm/ $^{\circ}\text{C}$

#### 9.2.2.2 Detailed Design Procedure

To get unipolar output from DACx1001, connect the negative reference input to ground as shown in Figure 59. The OPA189 is a zero-drift amplifier with  $\pm 0.02$  ppm/ $^{\circ}\text{C}$ . The DACx1001 has a temperature drift of offset error of  $\pm 0.04$  ppm/ $^{\circ}\text{C}$ . The temperature drifts of the DAC and amplifier might be neglected when compared to the temperature drift of the reference source. The best reference sources offer temperature drifts of the order of  $\pm 2.5$  ppm/ $^{\circ}\text{C}$  to  $\pm 3$  ppm/ $^{\circ}\text{C}$ . A temperature calibration is needed for the voltage reference to achieve the goal of  $\pm 2$  ppm/ $^{\circ}\text{C}$ .

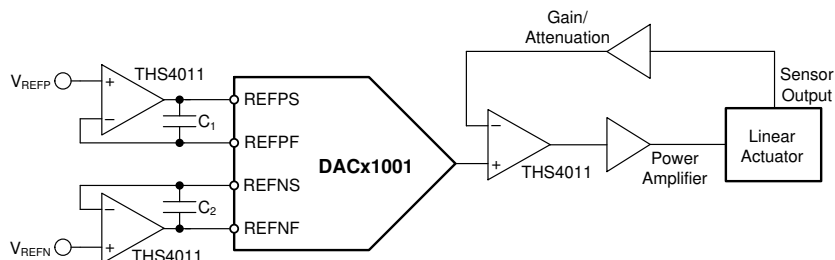
#### 9.2.2.3 Application Curves



## Typical Application (continued)

### 9.2.3 High-Precision Control Loop

High-precision control loops are used in precision motion-control applications, such as linear actuator control, servo motor control, galvanometer control, and more. The key requirements for such applications is resolution, monotonicity, settling time, and code-to-code glitch. [Figure 62](#) provides a simplified circuit of a linear actuator control circuit, wherein the DACx1001 commands the set point and an analog loop controls the actuator.



**Figure 62. High-Precision Control Loop**

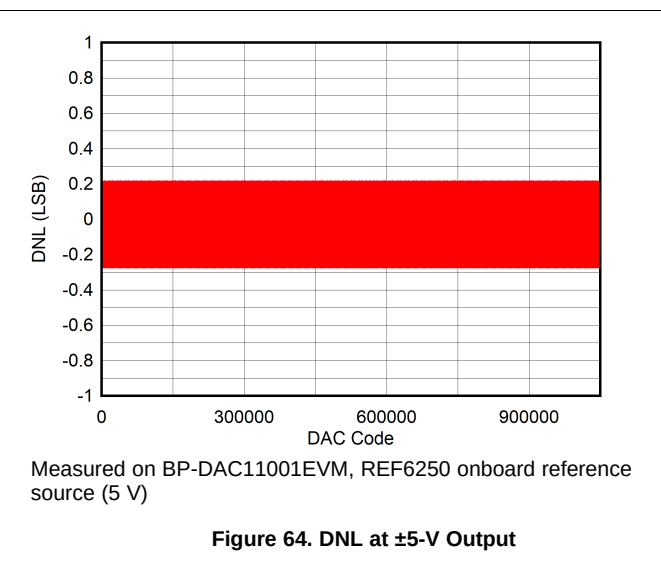
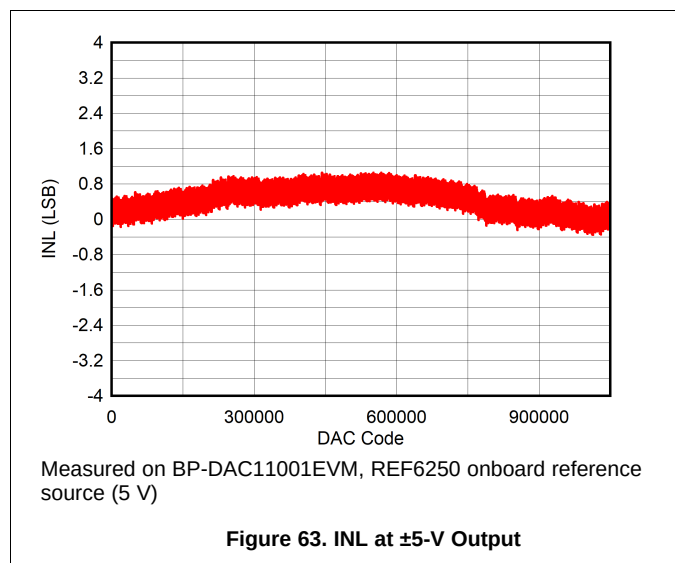
#### 9.2.3.1 Design Requirements

- DNL:  $\pm 1$  LSB max at 20-bits
- Settling time:  $< 2 \mu\text{s}$
- Code-to-code glitch:  $< 2 \text{ nV-s}$

#### 9.2.3.2 Detailed Design Procedure

The DACx1001 provides 20-bit monotonic resolution at  $< \pm 1$  LSB DNL. The device provides  $< 2 \mu\text{s}$  settling time and  $< 2 \text{ nV-s}$  code-to-code glitch for major carry transition. The reference and output buffer used for this design is the [THS4011](#), a high-speed amplifier with a 90-ns settling time. For the best settling response, use  $C_1$  and  $C_2$  between 10 pF to 50 pF.

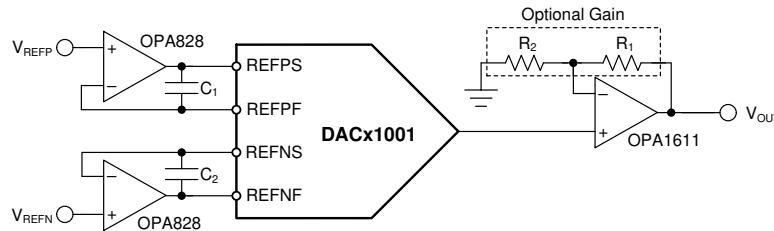
#### 9.2.3.3 Application Curves



## Typical Application (continued)

### 9.2.4 Arbitrary Waveform Generation (AWG)

Arbitrary waveform generation circuits are common in memory and semiconductor test equipment. These circuits are used to generate reference ac waveforms to test semiconductor devices. The key performance parameters of such circuits are THD, SNR, and the update rate. Figure 65 shows the basic building block example of an AWG circuit using the DACx1001.



**Figure 65. Arbitrary Waveform Generation**

#### 9.2.4.1 Design Requirements

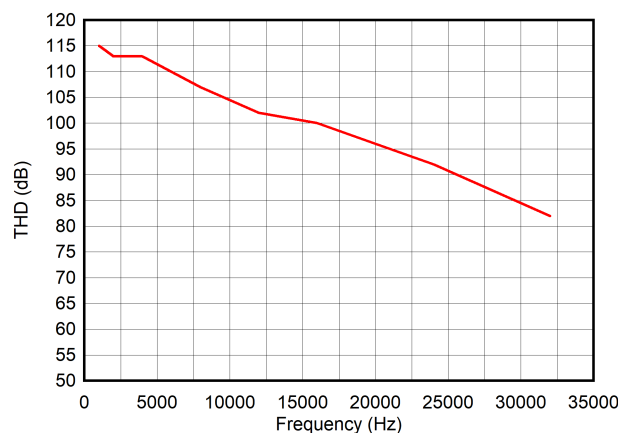
- THD at 1 kHz: > –105 dB
- Update rate: 100 kHz

#### 9.2.4.2 Detailed Design Procedure

The DACx1001 provides a THD of –105 dB at 1 kHz. The device provides update rates of up to 1 MHz, with marginal degradation in THD at higher frequencies. The OPA828 provides the best balance between the voltage and current noise densities, and is therefore an excellent choice to use as reference buffers. The OPA1611 is a low-distortion amplifier for high-THD applications.

#### 9.2.4.3 Application Curves

The test conditions for the THD values in the graph of Figure 66 are a  $\pm 3$ -V reference input on the BP-DAC11001EVM, and an external 3x gain at the DAC output. The THD calculation considers 11 harmonics; the even harmonics are omitted. When two DACs are used in a differential output mode, the even harmonics are cancelled to a large extent. Figure 66 shows an ideal scenario, when the even harmonics are completely cancelled out.



**Figure 66. THD vs Frequency**

## 9.3 System Examples

This section provides details on the digital interface and the embedded resistor configurations.

### 9.3.1 Interfacing to a Processor

The DACx1001 family of DACs works with a 4-wire SPI interface. The digital interface of the DACx1001 to a processor is shown in Figure 67. The DACx1001 has an  $\overline{\text{LDAC}}$  input option for synchronous output update. In ac-signal generation applications, the jitter in the  $\overline{\text{LDAC}}$  signal contributes to signal-to-noise ratio (SNR). Therefore, the  $\overline{\text{LDAC}}$  signal must be generated from a low-jitter timer in the processor. The CLR and ALARM pins are static signals, and therefore can be connected to general-purpose input-output (GPIO) pins on the processor. All active-low signals (SYNC,  $\overline{\text{LDAC}}$ , CLR, and ALARM) must be pulled up to IOVDD using 10-k $\Omega$  resistors. ALARM is an output pin from the DAC, so the corresponding GPIO on the processor must be configured as an input. Either poll the GPIO, or configured the GPIO as an interrupt to detect any failure alarm from the DAC. When using a high SCLK frequency, use source termination resistors, as shown in Interfacing to a Processor. Typically, 33- $\Omega$  resistors work on printed circuit boards (PCBs) with a 50- $\Omega$  trace impedance.

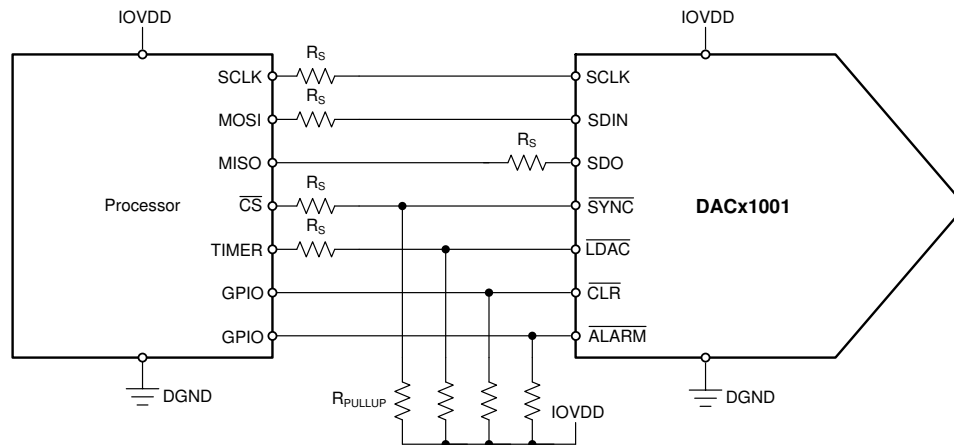


Figure 67. Interfacing to a Processor

### 9.3.2 Interfacing to a Low-Jitter $\overline{\text{LDAC}}$ Source

When the processor is not able to provide a low-jitter source for the  $\overline{\text{LDAC}}$  signal, an external low-jitter LDAC source can be used, as shown in Figure 68. The processor can take the  $\overline{\text{LDAC}}$  signal as an interrupt and trigger the SPI frame synchronously.

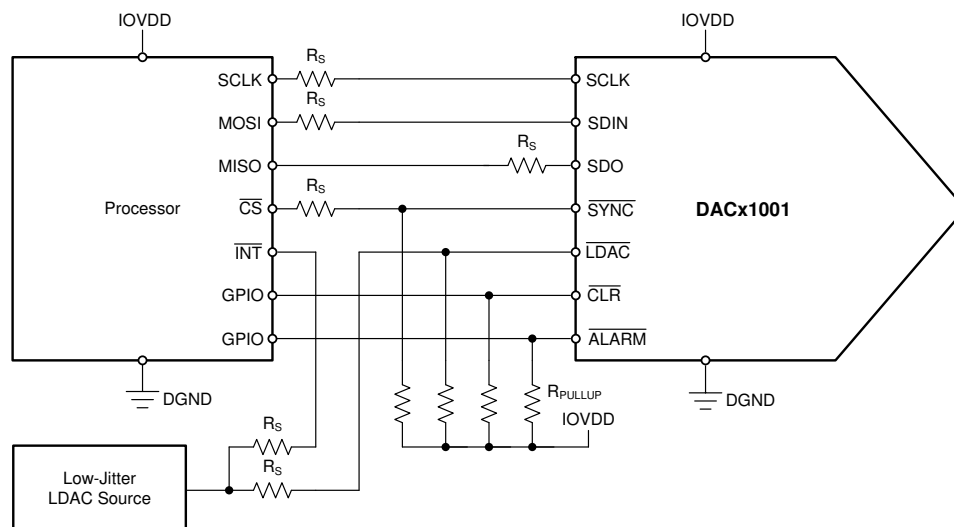


Figure 68. Interfacing to an External  $\overline{\text{LDAC}}$  Source

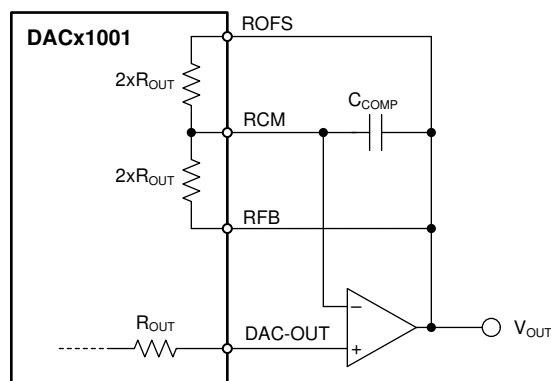
## System Examples (continued)

### 9.3.3 Embedded Resistor Configurations

The DACx1001 provides two embedded resistors with values is double the value of the output impedance of the R2R ladder. These resistors can be used in various configurations, as shown in the following subsections.

#### 9.3.3.1 Minimizing Bias Current Mismatch

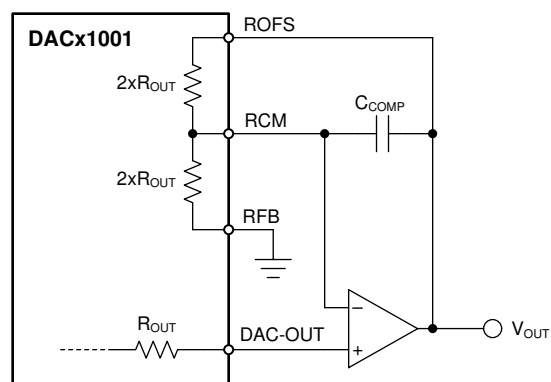
The bias current mismatch in the output amplifier can lead to offset error at the output. To minimize mismatch, the amplifier must have a matching resistor to that of the R2R output impedance on the feedback path. The feedback resistors are used in parallel for this purpose, as shown in [Figure 69](#). Some amplifiers may become unstable with a feedback resistor in the buffer configuration; therefore, a compensation capacitor ( $C_{COMP}$ ) might be needed, as shown. The typical value of this capacitor is in the range of 22 pF to 100 pF, depending on the amplifier.



**Figure 69. Minimizing Bias Current Mismatch**

#### 9.3.3.2 2x Gain configuration

The circuit of [Figure 69](#) can be configured for 2x gain by connecting one of the resistor ends to ground, as shown in [Figure 70](#).

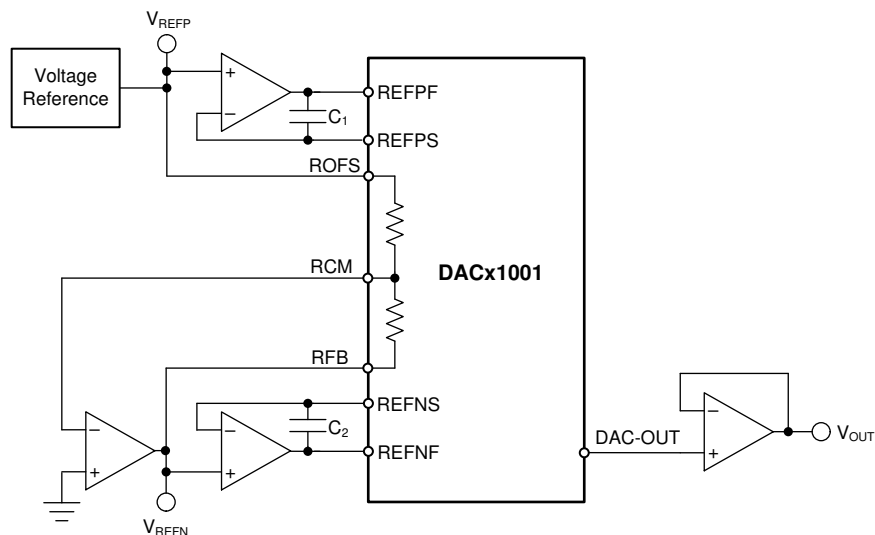


**Figure 70. 2x Gain Configuration**

## System Examples (continued)

### 9.3.3.3 Generating Negative Reference

Generating a negative reference is a challenge because of the fact that the circuit needs an inverting amplifier involving resistors. The resistor mismatch and temperature drift can lead to inaccuracy. The embedded, matched resistors in DACx1001 can be used as shown in [Figure 71](#), the inverting amplifier configuration, to generate an accurate negative reference voltage.



**Figure 71. Generating Negative Reference**

## 9.4 What to Do and What Not to Do

### 9.4.1 What to Do

- Follow recommended grounding, decoupling, and layout schemes for achieving best accuracy.
- Use a low-jitter LDAC source for best ac performance.
- Choose the appropriate amplifiers depending on the application requirements as explained in above sections.

### 9.4.2 What Not to Do

- Do not apply the reference before the DAC power supplies are powered on.
- Do not use the reference source directly with the DAC reference inputs without using buffers. or else the accuracy drastically degrades.

## 9.5 Initialization Set Up

The following text shows the pseudocode to get started with the DACx1001:

```
//SPI Settings
//Mode: Mode-1 (CPOL: 0, CPHA: 1)
//CS Type: Active Low, Per Packet
//Frame length: 32
//SYNTAX: WRITE <REGISTER (HEX ADDRESS)>, <HEX DATA>
//Select VREF, TnH mode (Good THD), LDAC mode and power-up the DAC
WRITE CONFIG (0x02), 0x004C80
//Write zero code to the DAC
WRITE DACDATA (0x01), 0x000000
//Write mid code to the DAC
WRITE DACDATA (0x01), 0x7FFFF0
//Write full code to the DAC
WRITE DACDATA (0x01), 0xFFFFF0
```

## 10 Power Supply Recommendations

To get the best performance out of the DACx1001, the power supply, grounding, and decoupling are very important. Use a PCB with a ground-plane reference, which helps in confining the digital return currents. A low mutual inductance path is created just beneath the high-frequency digital traces causing the return currents to follow the respective signal traces, thus minimizing crosstalk. On the other hand, dc signals spread over the ground plane without being confined below the signal trace. Therefore, in precision dc applications, limiting the common-impedance coupling is very difficult unless the ground planes are physically separated. Figure 72 shows a method to divide the grounds so that there is no common-mode current flow between the grounds, while maintaining the same dc potential across all grounds. This circuit assumes that the REFGND and LOAD-GND are provided from isolated power sources, therefore, there is no common-mode current flow through the reference or the load.

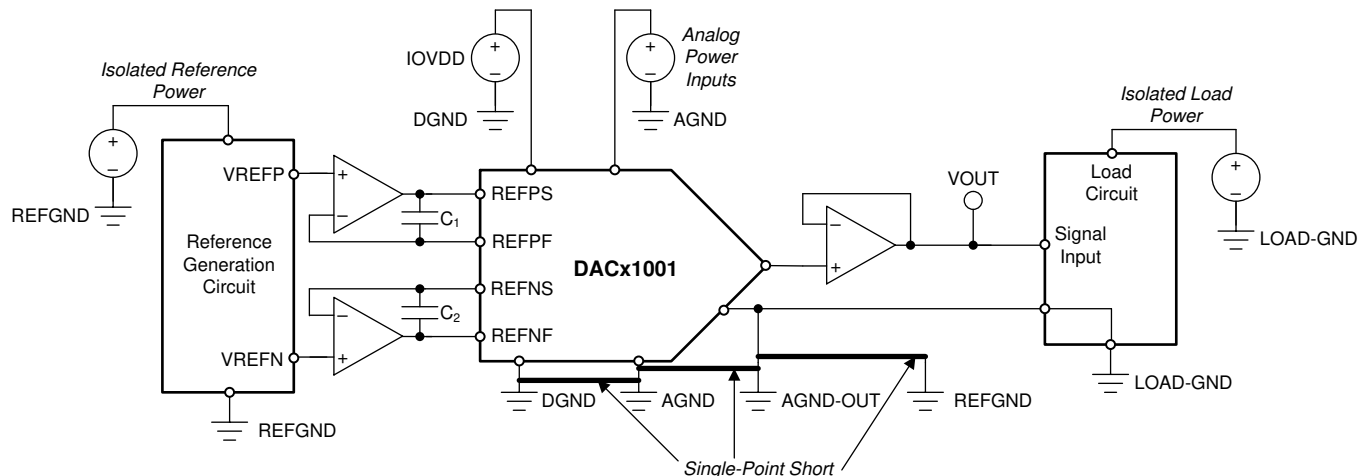


Figure 72. Power and Signal Grounding

When the load circuit is powered from a source referenced to AGND, and the LOAD-GND is shorted to AGND at the far end, the AGND-OUT must no longer be shorted to AGND locally near the DAC. The local shorting creates a ground loop, otherwise. The resulting connection that avoids the ground loop is shown in Figure 73.

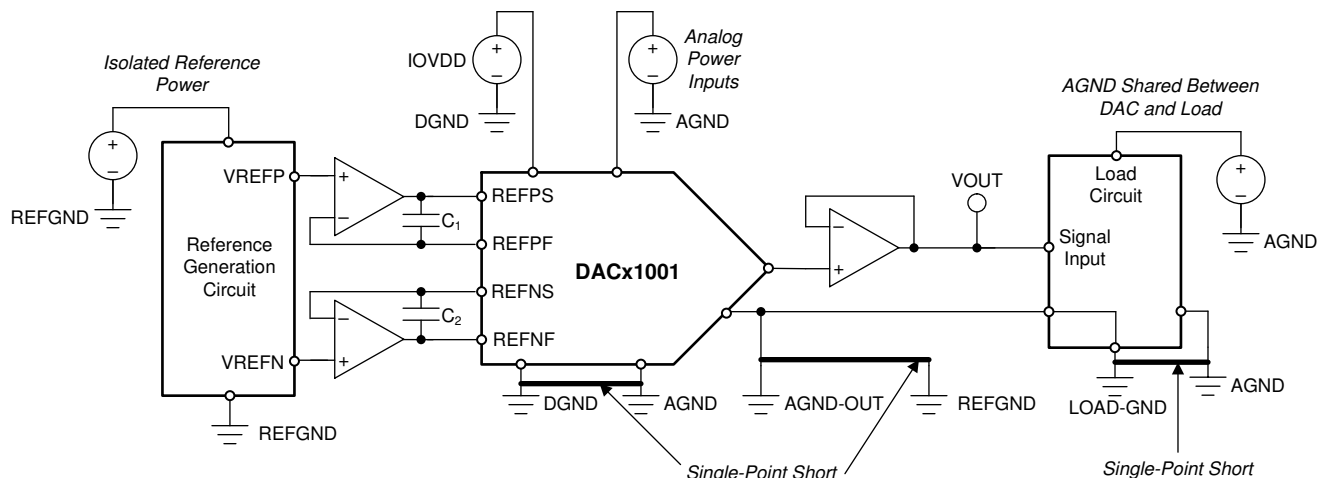
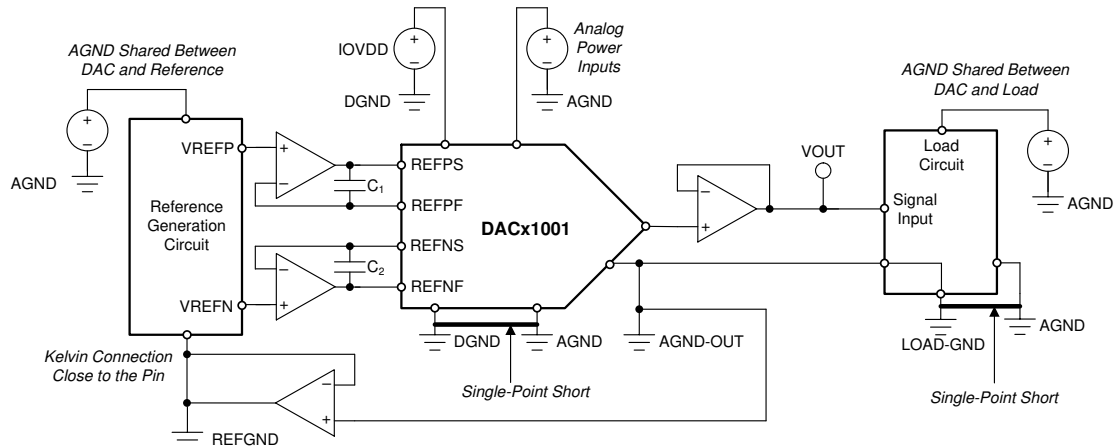


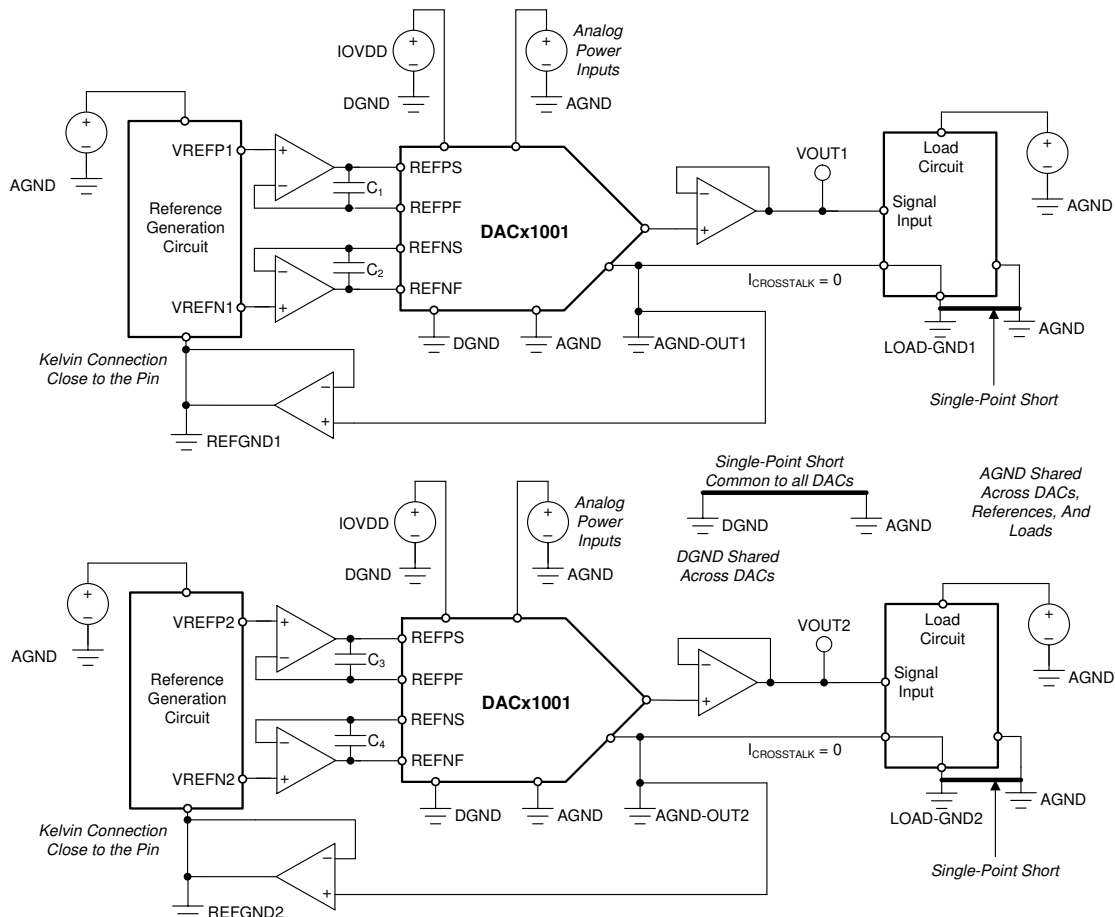
Figure 73. Grounding Scheme When AGND is Load Ground

When the reference source is powered from a power source with AGND as the ground, there is a possibility of common-impedance coupling causing a code-dependent shift in the reference voltage. To avoid undesired coupling, drive REFGND using a buffer that maintains the reference ground potential equals to that of AGND-OUT, as shown in Figure 74.



**Figure 74. Connecting the Reference Ground**

Channel-to-channel dc crosstalk is a major concern in multichannel applications, such as battery test equipment. While the DACx1001 is single-channel, the crosstalk problem can appear at a system level when using multiple DACx1001 devices. The problem becomes severe when the grounds of the loads are shorted together creating a possible ground loop. In such cases, avoid the local short between AGND and AGND-OUT. Use a single short between AGND and DGND for all the DACs. If the PCB layout allows for the digital signal and analog power supplies to be kept separate, DGND and AGND can be combined to a single ground plane. [Figure 75](#) shows an example circuit for minimizing dc crosstalk across DAC channels in a system.

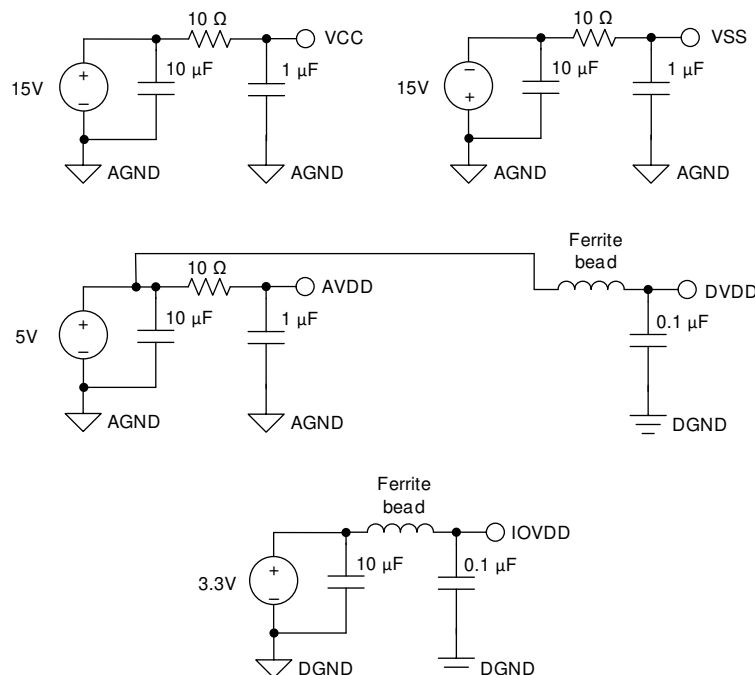


**Figure 75. Minimizing Multichannel DC Crosstalk**

Power-supply bypassing and decoupling is key to keeping power supply noise, switching transients, and common-mode currents away from the DAC output. There are three main objective of power-supply bypassing:

- *Filtering*: Filter out noise and ripple from power supplies
- *Bypassing*: Supply switching or load transient currents locally by avoiding trace inductances
- *Decoupling*: Stop local transient currents from impacting other circuits

To achieve these objectives, use the following 3-element scheme. Place a decoupling capacitor close to every power supply pin to provide the local current path for load and circuit switching transients. This capacitor must be referenced to the respective load ground for best load transient suppression. Use a 0.1- $\mu\text{F}$  to 1- $\mu\text{F}$ , X7R, multilayer ceramic capacitor (MLCC) for this purpose. For analog power supplies, a 10- $\Omega$  series resistor provides the best decoupling. For filtering the power-supply noise and ripple, 10- $\mu\text{F}$  capacitors work best when placed at the power entry point of the board. An example decoupling scheme is shown in [Figure 76](#).



**Figure 76. Power-Supply Decoupling**

## 10.1 Power-Supply Sequencing

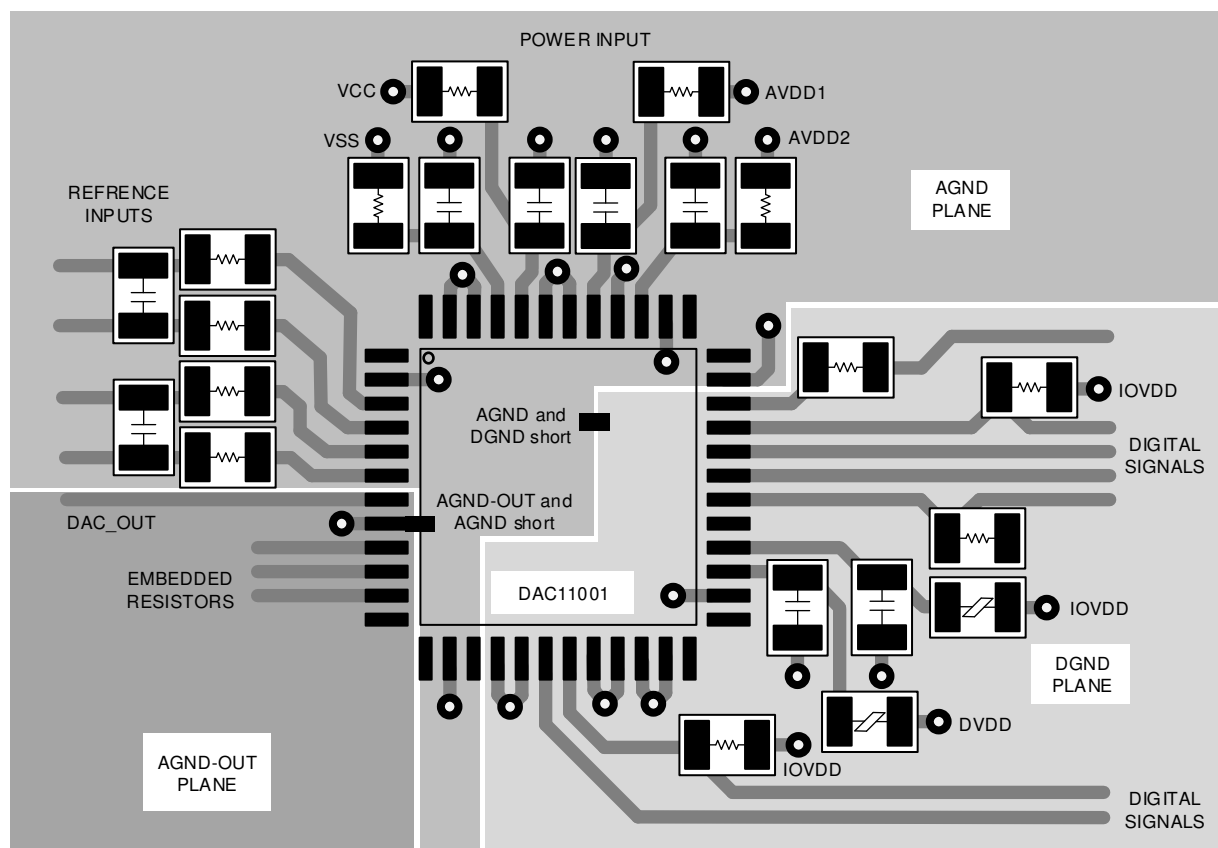
The DACx1001 do not require any power-supply sequence. However, the power supplies to the AVDD pin must be capable of providing 30-mA of current if  $V_{SS}$  ramps before  $AV_{DD}$ . This current is derived from the AVDD pin, and flows out of the VSS pin. This condition is transient, and the device stops consuming this current when the power supplies are ramped up. To avoid this condition, make sure to ramp  $AV_{DD}$  before  $V_{SS}$ .

## 11 Layout

### 11.1 Layout Guidelines

PCB layout plays a significant role for achieving desired ac and dc performance from the DACx1001. The DACx1001 has a pinout that supports easy splitting of the noisy and quiet grounds. The digital signals are available on two adjacent sides of the device; whereas, the power and analog signals are available separate sides. [Figure 77](#) shows an example layout, where the different ground planes have been clearly demarcated. The figure also shows the best positions for the single-point shorts between the ground planes. For best power-supply bypassing, place the bypass capacitors close to the respective power pins as shown. Provide unbroken ground reference planes for the digital signal traces, especially for the SPI and LDAC signals.

### 11.2 Layout Example



**Figure 77. Layout Example**

## 12 Device and Documentation Support

### 12.1 Device Support

#### 12.1.1 Development Support

[BP-DAC11001 Evaluation Module](#)

### 12.2 Documentation Support

#### 12.2.1 Related Documentation

For related documentation see the following:

- Texas Instruments, [BP-DAC11001EVM user's guide](#)
- Texas Instruments, [Impact of Code-to-Code Glitch in Precision Applications application brief](#)

### 12.3 Related Links

[Table 12](#) lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to order now.

**Table 12. Related Links**

| PARTS     | PRODUCT FOLDER             | ORDER NOW                  | TECHNICAL DOCUMENTS        | TOOLS & SOFTWARE           | SUPPORT & COMMUNITY        |
|-----------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| DAC11001A | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| DAC91001  | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| DAC81001  | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |

### 12.4 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. In the upper right corner, click on *Alert me* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

### 12.5 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 12.8 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

## 13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

## PACKAGING INFORMATION

| Orderable part number         | Status<br>(1) | Material type<br>(2) | Package   Pins  | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|-------------------------------|---------------|----------------------|-----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">DAC11001APFBR</a> | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC11001A           |
| DAC11001APFBR.B               | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC11001A           |
| <a href="#">DAC11001APFBT</a> | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC11001A           |
| DAC11001APFBT.B               | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC11001A           |
| <a href="#">DAC81001PFBR</a>  | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC81001            |
| DAC81001PFBR.B                | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC81001            |
| <a href="#">DAC81001PFBT</a>  | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC81001            |
| DAC81001PFBT.B                | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC81001            |
| <a href="#">DAC91001PFBR</a>  | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC91001            |
| DAC91001PFBR.B                | Active        | Production           | TQFP (PFB)   48 | 1000   LARGE T&R      | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC91001            |
| <a href="#">DAC91001PFBT</a>  | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC91001            |
| DAC91001PFBT.B                | Active        | Production           | TQFP (PFB)   48 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-3-260C-168 HR               | -40 to 125   | DAC91001            |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

<sup>(4)</sup> **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

<sup>(5)</sup> **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

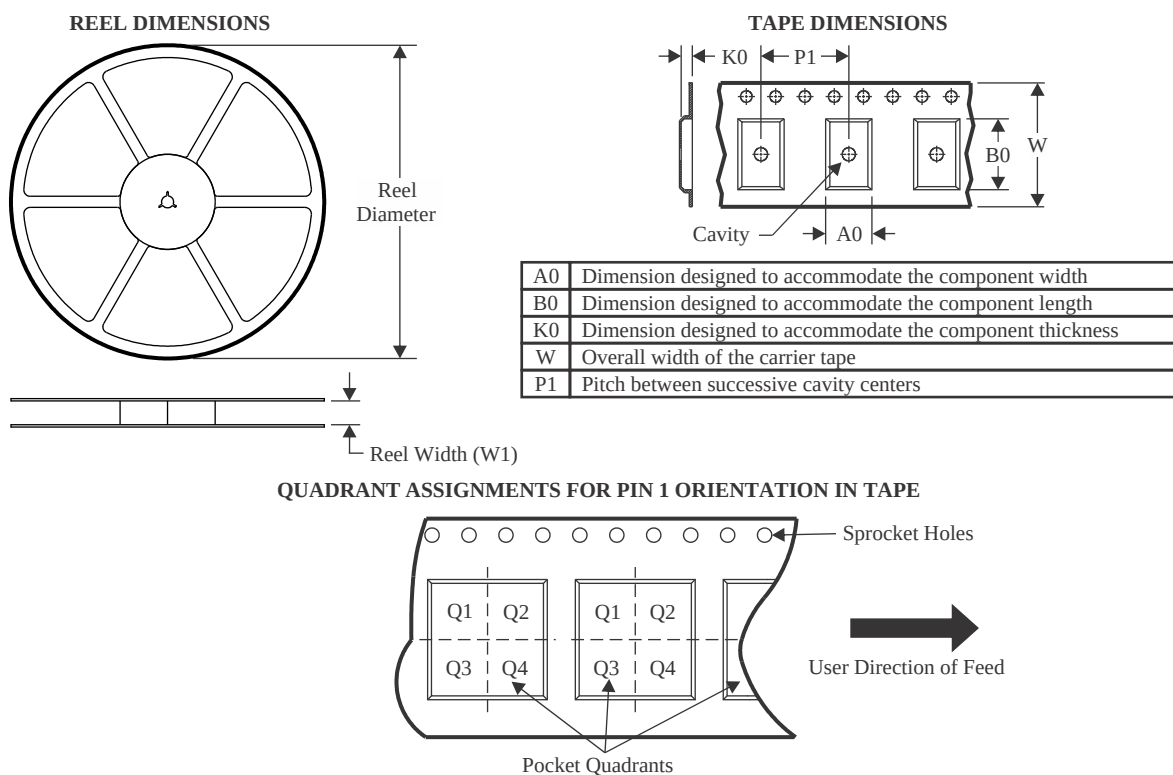
<sup>(6)</sup> **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "-" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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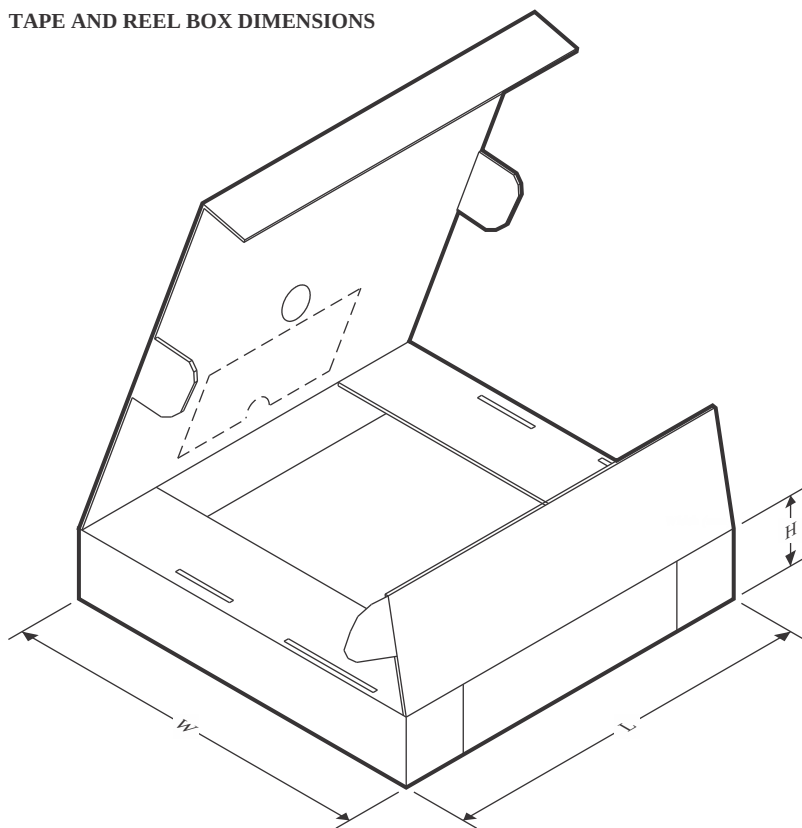
## TAPE AND REEL INFORMATION



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| DAC11001APFBR | TQFP         | PFB             | 48   | 1000 | 330.0              | 16.4               | 9.6     | 9.6     | 1.5     | 12.0    | 16.0   | Q2            |
| DAC81001PFBR  | TQFP         | PFB             | 48   | 1000 | 330.0              | 16.4               | 9.6     | 9.6     | 1.5     | 12.0    | 16.0   | Q2            |
| DAC91001PFBR  | TQFP         | PFB             | 48   | 1000 | 330.0              | 16.4               | 9.6     | 9.6     | 1.5     | 12.0    | 16.0   | Q2            |

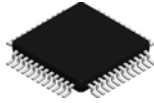
## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| DAC11001APFBR | TQFP         | PFB             | 48   | 1000 | 350.0       | 350.0      | 43.0        |
| DAC81001PFBR  | TQFP         | PFB             | 48   | 1000 | 350.0       | 350.0      | 43.0        |
| DAC91001PFBR  | TQFP         | PFB             | 48   | 1000 | 350.0       | 350.0      | 43.0        |

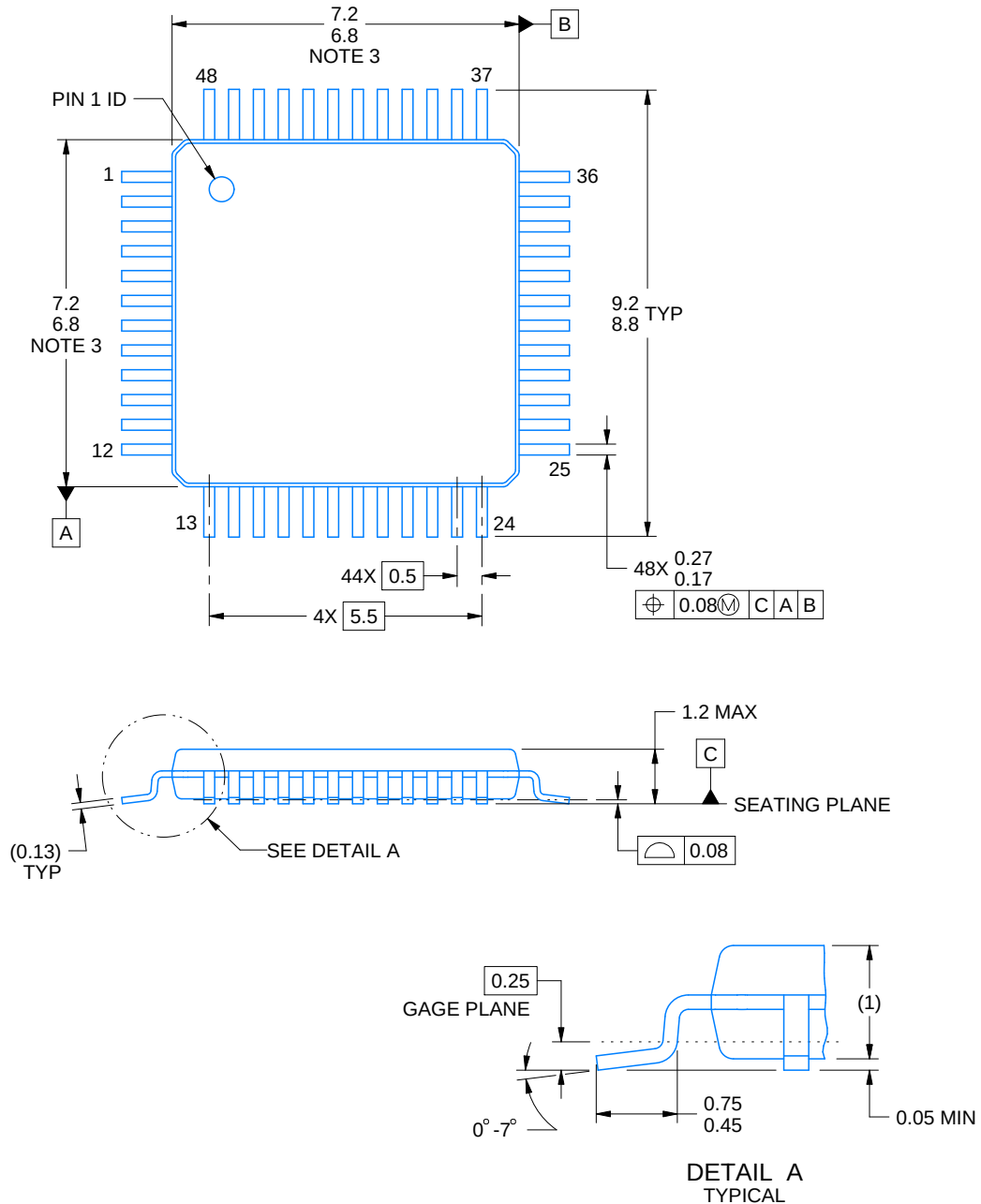
PFB0048A



# PACKAGE OUTLINE

TQFP - 1.2 mm max height

PLASTIC QUAD FLATPACK



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## NOTES:

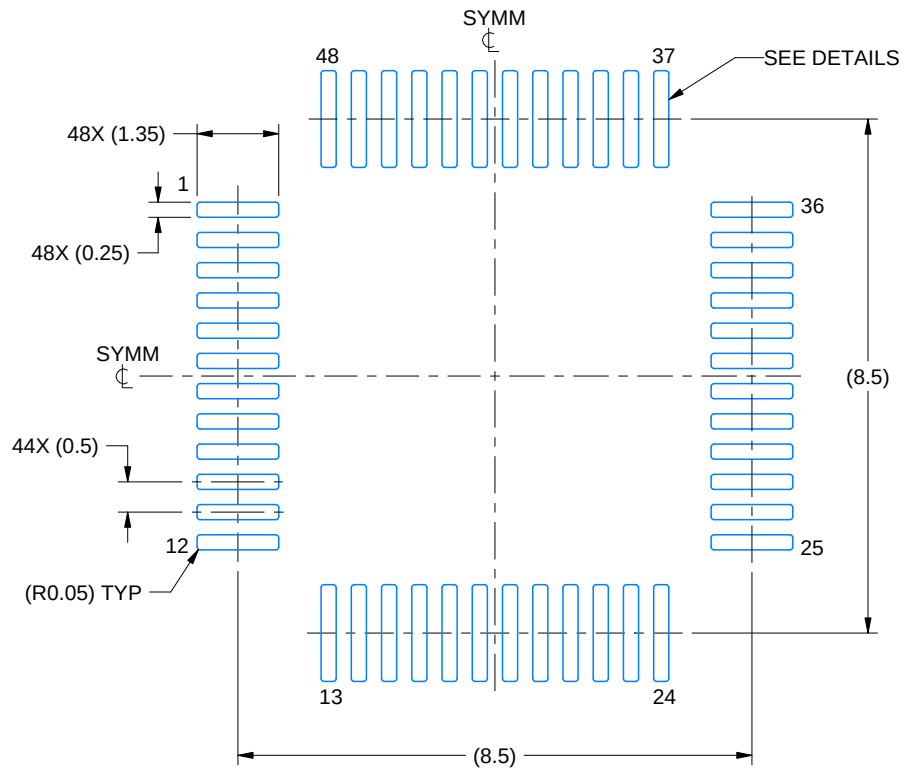
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC registration MS-026.

# EXAMPLE BOARD LAYOUT

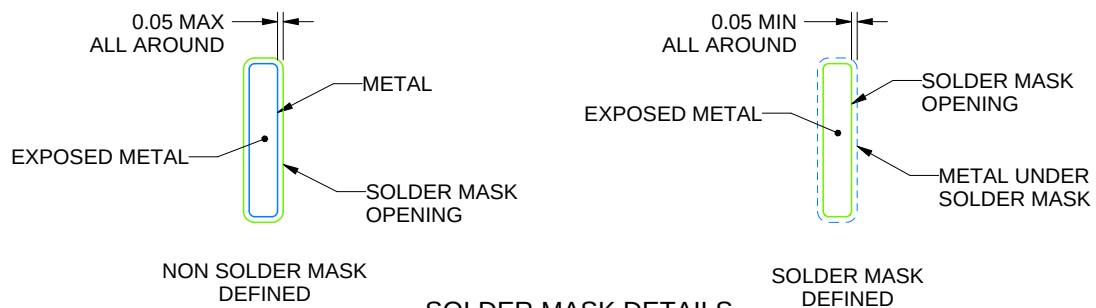
PFB0048A

TQFP - 1.2 mm max height

PLASTIC QUAD FLATPACK



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:8X



SOLDER MASK DETAILS

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NOTES: (continued)

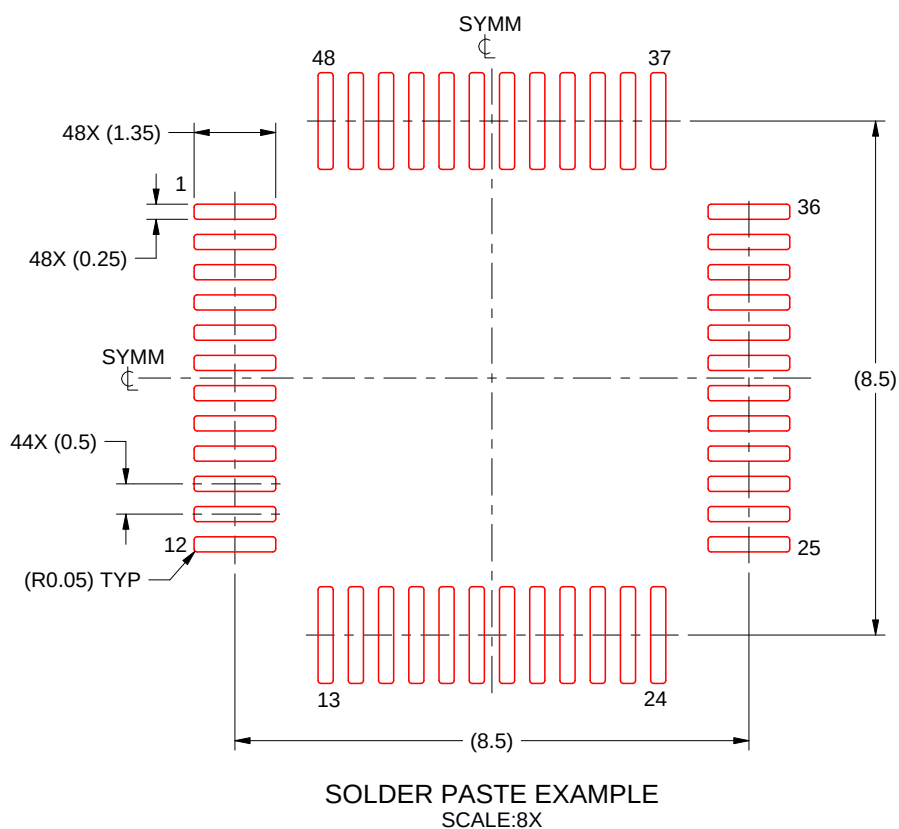
4. Publication IPC-7351 may have alternate designs.
5. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

PFB0048A

TQFP - 1.2 mm max height

PLASTIC QUAD FLATPACK



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NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
7. Board assembly site may have different recommendations for stencil design.

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